



Mechanism of germanium enrichment in the world-class Huize MVT Pb–Zn deposit, southwestern China

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Abstract

The world-class Huize deposit hosts significant germanium (Ge) resources in the Sichuan–Yunnan–Guizhou (SYG) Mississippi Valley-type (MVT) Pb–Zn province of China. The distribution and enrichment mechanism of Ge is still poorly understood. In the main ore-forming stage of Huize, we identified six sphalerite colors from C1 (black) to C6 (white) in transmitted light. Two color sequences are confirmed, including C1 → C2 → C3 → C6 and C1 → C2 → C4 → C5 → C6. We used multiple analytical methods to reveal the Ge distribution and incorporation mechanism into sphalerite and the possible enrichment factors. Our results show that Ge occurs as argutite (GeO₂), and in the sphalerite crystal lattice, C1 and C3 sphalerite has up to 593 ppm Ge. Two substitution mechanisms, i.e., $\text{Ge}^{4+} + \square_{(\text{vacancy})} \rightarrow 2\text{Zn}^{2+}$ (e.g., C1 and C2) and $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 3\text{Zn}^{2+}$ (e.g., C2, C3, C4, and C5), are inferred from the Huize sphalerite. They show different spatial structures of sphalerite and a weak shift of the white line observed by high-resolution X-ray absorption near-edge structure (XANES) spectroscopy. The trace-element composition of sphalerite suggests that reduced sulfur content of the ore-forming fluid contributes to Ge enrichment, followed by high temperature (> 300 °C).

Keywords Germanium · EBSD · HR-synchrotron · Colored sphalerite · Huize deposit

Introduction

Germanium (Ge) is crucial for many high-tech fields (optical fiber, infrared optics, electronics, and solar devices) and is often classified as a critical metal (Shanks et al. 2017; Jiang et al. 2023). Despite its relatively low abundance in the continental crust (~ 1.6 ppm), Ge can be enriched in

economic mineralization due to siderophile, lithophile, chalcophile, and organophile properties (Bernstein 1985; Taylor and McLennan 1995; Höll et al. 2007; Rosenberg 2008; Frenzel et al. 2014; Zheng et al. 2023). Actually, the global Ge supply mainly comes from Pb–Zn deposits and coals, and high-grade Ge-rich Pb–Zn deposits are of great economic significance because of their advantages of easy mining and low pollution (Ma et al. 2019; U.S. Geological Survey 2022; Zheng et al. 2023). In these Ge-rich Pb–Zn deposits, Ge may substitute for Zn in the sphalerite crystal lattice, for example: (1) $\text{Ge}^{2+} \leftrightarrow \text{Zn}^{2+}$, $\text{Ge}^{4+} + \square \leftrightarrow 2\text{Zn}^{2+}$, $\text{Ge}^{4+} + 2\text{Cu}^+/\text{Ag}^+ \leftrightarrow 3\text{Zn}^{2+}$, and $\text{Ge}^{4+} + 2\text{Fe}^{2+} \leftrightarrow 4\text{Zn}^{2+}$ (Cook et al. 2009; Belissont et al. 2014; Zhuang et al. 2019; Yuan et al. 2018; Cugerone et al. 2021), and (2) $4(\text{Cu}^+ + \text{Sb}^{3+}) + (\text{Ge}^{4+} + 2\text{Ag}^+) + 2\square \leftrightarrow 13\text{Zn}^{2+}$ and $(\text{Cu}^+ + \text{Ag}^+) + (\text{Sb}^{3+} + \text{In}^{3+}) + (\text{Sn}^{4+} + \text{Ge}^{4+}) \leftrightarrow 8\text{Zn}^{2+}$ (Wei et al. 2018; Li et al. 2020). Synchrotron experiments confirmed that Ge and Cu occur as Ge^{4+} and Cu^+ in sphalerite, further constraining the incorporation mechanism of Ge into sphalerite (e.g., Belissont et al. 2014; Cook et al. 2015). The occurrence of Ge^{2+} in sphalerite was only reported from the MVT deposits in Tennessee, USA (Bonnet et al. 2017).

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In sphalerite, Ge content is frequently zoned but no specific color in transmitted light microscopy for Ge-rich zones is generally observed (e.g., Gagnevin et al. 2014; Belissont et al. 2014; Bauer et al. 2019; Niu et al. 2023). Ge enrichment in sphalerite is thought to be primarily controlled by fluid temperature (Frenzel et al. 2016; Liu et al. 2020), but the redox state of the fluid during sphalerite precipitation could be an influencing factor (Pokrovski et al. 2000). Metamorphism can result in Ge redistribution in the parent sphalerite crystal lattice to Ge-bearing independent minerals (Cugerone et al. 2019). The mechanism for the super-enrichment of Ge and other critical metals has attracted much attention (Luo et al. 2022; Jiang et al. 2023; Zheng et al. 2023).

The Huize deposit is the largest Pb–Zn deposit in the SYG area of southwestern China. It contains more than 7 Mt Pb + Zn (grading at 25–35%) and 800 t Ge (grading at 30–100 g/t) metal reserves (Huang et al. 2004; Zhang et al. 2013; Bao et al. 2017; Leach and Song 2019; Tao et al. 2019). The Huize deposit is representative of high-grade Ge-rich Pb–Zn deposits and is classified as the “Huize-type” MVT deposit, characterized by higher Pb–Zn grade than most of the world’s MVT Pb–Zn deposits (3–10%; Leach et al. 2010) and large Ge resources (Huang et al. 2004; Bao et al. 2017; Han et al. 2019). The lack of detailed research on sphalerite type and related Ge distribution prevents the study of the enrichment mechanism of Ge in the Huize Pb–Zn deposit.

Many Ge-bearing minerals are known, however, few of them have been reported in Pb–Zn deposits, for example, argutite (GeO_2), germanite ($\text{Cu}_{13}\text{Fe}_2\text{Ge}_2\text{S}_{16}$), briarite [$\text{Cu}_2(\text{Fe}, \text{Zn})\text{GeS}_4$], carboirite [$\text{FeA}_{12}\text{GeO}_5(\text{OH})_2$], brunogeierite (Fe_2GeO_4), and stottite [$\text{FeGe}(\text{OH})_6$] in the metamorphosed Pyrenean deposits (France), Tsumeb (Namibia), and Kipushi (D.R. Congo) deposits (Bernstein 1985; Cugerone et al. 2018, 2019). Currently, Ge-bearing independent minerals are yet to be found in the Huize deposit, except for one report on suspected krieselite [$\text{Al}_2(\text{GeO}_4)\text{F}_2$] without detailed mineral composition (i.e., Zhang et al. 2008). Based on extensive laser ablation inductively coupled plasma mass spectroscopy (LA-ICP-MS) data, flat time-integrated signals indicate that sphalerite in the Huize deposit may be the primary host mineral for Ge and contains up to 930 ppm Ge (Liu et al. 2022). Element substitutions of $\text{Ge}^{2+} + 2\text{Cu}^+ \rightarrow 2\text{Zn}^{2+}$ and $\text{Ge}^{2+} \rightarrow \text{Zn}^{2+}$ were speculated according to correlation patterns (e.g., Ye et al. 2011; Ji 2019; Oyebamiji et al. 2023).

This study focuses on the following scientific questions on the Huize deposit: (1) Is Ge only hosted in sphalerite crystal lattice? (2) What is the distribution of Ge in the different sphalerite generations? (3) What is the enrichment mechanism of Ge in sphalerite? (4) Can later geological events redistribute the Ge concentration, as mentioned

by Han et al. (2007) and Cugerone et al. (2019)? (5) Why does the Huize deposit host more Ge resource than most other MVT deposits in the world? First, we use electron backscattered diffraction (EBSD) to identify Ge-bearing independent minerals, sphalerite structure, and potential intragranular deformation. Second, micro X-ray fluorescence (μ -XRF), synchrotron micro-focused X-ray fluorescence (SXRF), and LA-ICP-MS mappings are applied to the macroscopic (whole-section) distribution of major elements and microscopic (local-section) trace element distribution in sphalerite, respectively. Furthermore, electron probe microanalysis (EPMA) and LA-ICP-MS spot analysis are used for accurately confirming the sphalerite composition, and high-resolution synchrotron XANES is applied to verify spatial structural changes in sphalerite for different element substitutions.

Regional geology

The South China Craton is bordered to the north by the North China Craton along the Qinling-Dabie orogenic belt and is adjacent to the Tibetan tectonic domain in the west (Fig. 1A; Meng and Zhang 2000). The NE-trending Jiangnan orogenic belt divides the South Craton into the Yangtze Block to the northwest and the Cathaysia Block to the southeast (Fig. 1A). In the western part of the Yangtze Block, a huge mantle plume triggered the ca. 260 Ma Emeishan basalt province of 250,000 km² (Fig. 1A; Bai et al. 2014; Wang et al. 2020; Tang et al. 2021), and the Triassic granitic intrusions from 255 to 200 Ma are widespread in the Cathaysia Block (Fig. 1B; Yu et al. 2005; Hu et al. 2017).

The Huize deposit is situated at the triangular junction of the SYG area, southwestern Yangtze Block (Fig. 1B and C). The strata of the SYG area comprise a metamorphic basement and an overlying sedimentary cover (Bao et al. 2017). The metamorphic basement comprises the Neoproterozoic Kangding Group (crystalline basement) and Meso-Neoproterozoic Kunyang and Huili Groups (fold basement; Zhu et al. 2020). The Kangding Group is a set of dynamic metamorphic rocks composed of plagioclase gneiss, migmatitic gneiss, amphibolite, and minor granulite. The Kunyang and Huili Groups are dominated by marine sedimentary rocks and volcanic tuffs, which underwent medium–low-grade metamorphism at approximately 750 Ma (Liu et al. 2017). The overlying sedimentary cover comprises the Neoproterozoic Ediacaran to Mesozoic Triassic marine and subsequent continental sedimentary sequences (Zhang et al. 2015). The strata from the Ediacaran to Cambrian comprise pyroclastic rocks and clastic sedimentary rocks, and the Ordovician to Silurian sedimentary rocks contain limestone, dolomite, sandstone, and shale (Xu et al. 2020). Dolomitic

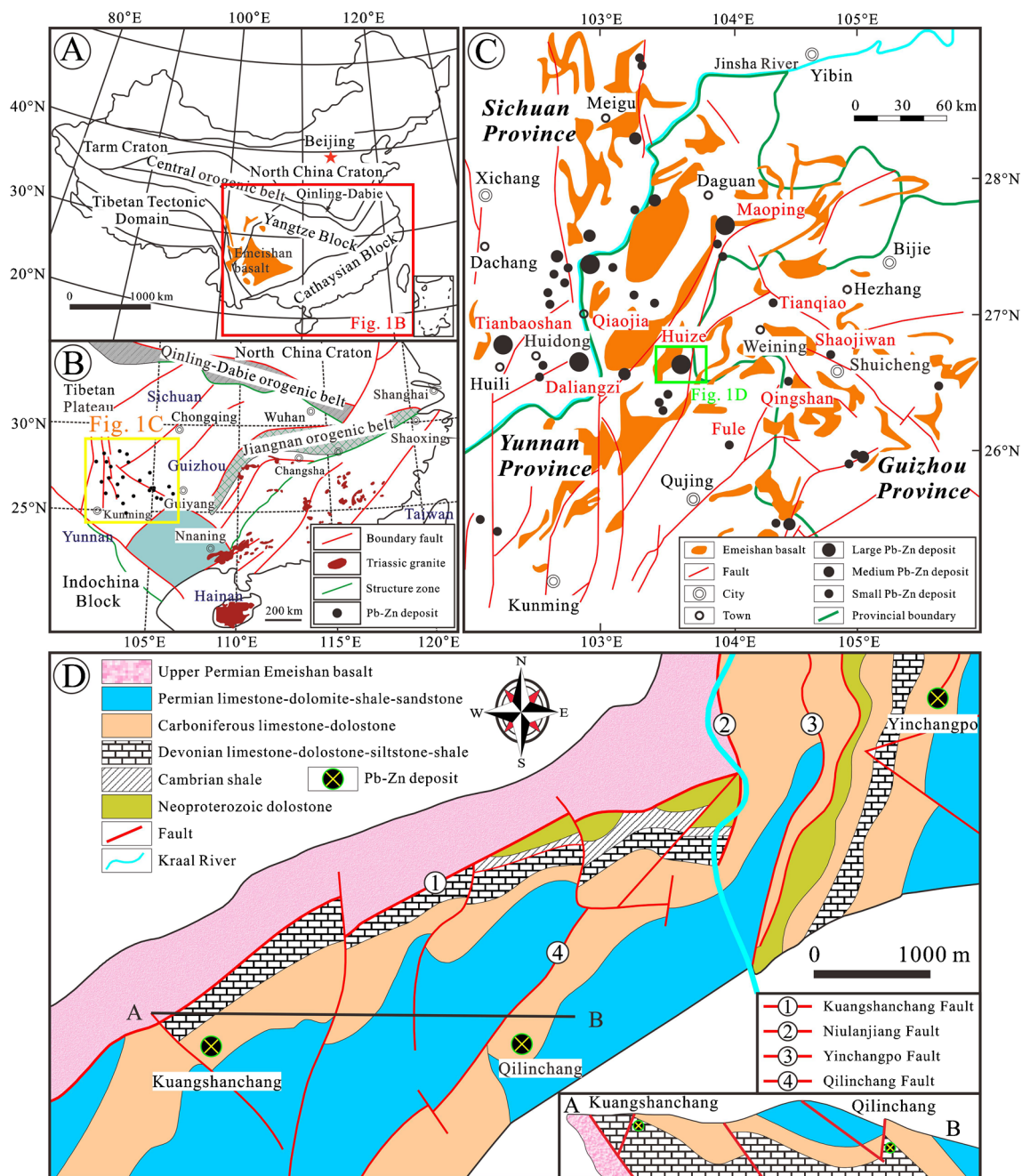


Fig. 1 **A** Tectonic framework of China, showing the distribution of the Emeishan basalt province (after Bao et al. (2017)); **B** simplified geological map of the South China Craton and internal tectonic pattern (after Hu et al. 2017); **C** geological map of the Sichuan–Yunnan–

Guizhou (SYG) area, showing major Pb–Zn deposits (after Meng et al. (2019)); **D** geological map of the Huize Pb–Zn deposit, showing the three primary mining areas of Qilinchang, Kuangshanchang, and Yinchangpo (after Han et al. 2007)

limestone, dolomite, quartz sandstone, and calcareous sandstone dominate the Devonian to Carboniferous units, while the Permian rocks comprise dolomitic limestone, breccia limestone, and argillaceous siltstone (Kong et al. 2018). Since the Jurassic period, many types of continental sediments have been deposited in the SYG area,

consisting of sandstone, conglomerate, and mud rocks (Wu et al. 2013).

There are fifteen major faults along the near NS, NE, and NW directions, constituting the SYG area's tectonic framework (Han et al. 2015). The near-NS trending faults originated as extensional features during the Proterozoic

era but later transformed into compressive and torsional faults in the Early Mesozoic period (Fig. 1C). The near NE- and NW-trending faults are primarily extensional and compressive, respectively, pre-dating the intrusion of Emeishan basalt into the region (Han et al. 2019). The distribution of Emeishan basalt within the SYG area spans a considerable thickness of up to 800 m, gradually thinning from west to east (Fig. 1C).

More than 400 Pb–Zn deposits (occurrences) are hosted in the SYG area, including a dozen large and giant Pb–Zn deposits such as Huize, Maoping, Daliangzi, and Tianbaoshan, as well as numerous medium–small Pb–Zn deposits like Fule, Tianqiao, Shaojiwan, and Qingshan (Fig. 1C; Hu and Zhou 2012; Zhou et al. 2014). Most of the Pb–Zn deposits in the SYG area are formed in 228–191 Ma and have no genetic link to the Emeishan basalt province (Yang et al. 2019 and references therein). Mixing of metal-bearing basin brine and reduced sulfur-bearing fluid is widely accepted as the ore-forming mechanism of these deposits (e.g., Huang et al. 2004; Li et al. 2006; Wang et al. 2016; Meng et al. 2019; Wu et al. 2021; Xiang et al. 2020).

Deposit geology

The Huize includes three mining areas: major Qilinchang and Kuangshanchang and minor Yinchangpo (Fig. 1D). The Qilinchang and Kuangshanchang mining areas are both concentrated in the Devonian limestone-dolostone-siltstone-shale of the fractured footwall (Fig. 1D). The Qilinchang and Kuangshanchang contain both sulfide and oxide ore bodies, with industrial production currently focused on extracting sulfide ore bodies dominated by sphalerite and galena (Fig. 2A and B; Han et al. 2019). Due to the extensive groundwater action, the oxide ore body was likely converted from the sulfide ore body. All samples collected in this study were obtained from fresh sulfide ore bodies within the Qilinchang and Kuangshanchang mining areas (Fig. 2A and B).

The strata exposed in the deposit include a metamorphic basement of the Mesoproterozoic Kunyang Group and a sedimentary cover ranging from the late Neoproterozoic to Mesozoic. The sedimentary cover comprises the Ediacaran Dengying Group, Cambrian Qiongzhusi Group, Devonian Zaige and Haikou Groups, Carboniferous Mapping, Weining,

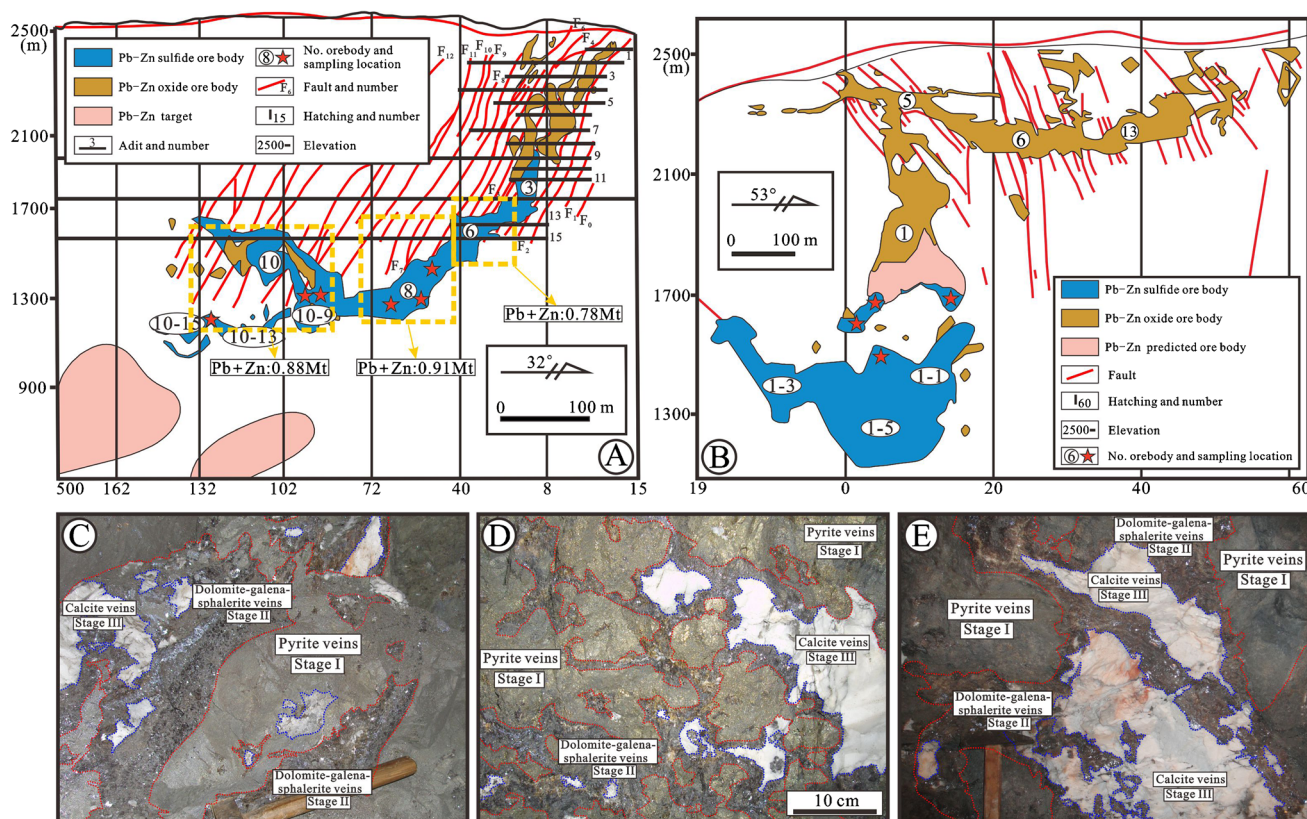


Fig. 2 **A** Section projection of Pb–Zn ore body and target in the Qilinchang mining area (after Han et al. (2019)); **B** vertical section of Pb–Zn (predicted) ore body in the Kuangshanchang mining area (after Han et al. (2019)); **C–E** downhole photos of the Huize Pb–Zn

deposit showing three distinct hydrothermal stages: (1) dolomite–pyrite veins in stage I, (2) dolomite–galena–sphalerite veins in stage II, and (3) pure calcite veins in stage III

Baizuo and Datang Groups, and Permian Qixia-Maokou and Liangshan Groups (Supplementary Fig. 1). Only localized occurrences of coal seams and carbonaceous shales are found within the Qiongzhusi and Liangshan Groups. The Baizuo Group represents the primary ore-bearing stratum of this deposit, exhibiting a thickness of 40–60 m, which is in conformable contact with the underlying Datang Group. The lower part of the Baizuo Group comprises light-gray dolomitic and bioclastic limestones interbedded with evaporites, with a small quantity of coral and brachiopod fossils in the bioclastic limestones. In the middle and upper part of the Baizuo Group, there are a series of pale yellow and gray middle-coarse-grained dolomites and light-gray dolomitic limestones with minor bioclastic limestones. A significant expanse of the Emeishan basalt is present on the northwest side of this deposit, with a total thickness of 600–800 m (Li et al. 2006). An unconformable contact can be observed between the Emeishan basalt and the underlying Qixia–Maokou Group through the primary NEE-trending fault (Fig. 1D).

A series of NW, NE, NNW, near SN, and near EW-trending faults are reported in the Huize area. Among them, the NE–SW-trending compressive faults (Kuangshanchang, Qilinchang, Niulanjiang, and Yinchangpo) are the primary ore-controlling structures in the Huize Pb–Zn deposit and are intersected by late NW-trending normal faults (Fig. 1D; Ren et al. 2018). The Kuangshanchang, Qilinchang, and Yinchangpo faults are considered early compared to the Pb–Zn ore emplacement (Fig. 1D; Han et al. 2019).

The Qilinchang fault extends approximately 32 km in a direction of 20–30° NE (with an eastward dip of 50–60°) and intersects with the Kuangshanchang fault in the northeast Kraal River (Fig. 1D). The Qilinchang mining area encompasses a total of 77 Pb–Zn ore bodies (20 economic ore bodies), and the Pb–Zn metal reserves are approximately 3.32 Mt, with average grades of 17.5% Zn and 6.6% Pb (Zhou et al. 2001). Most Pb–Zn ore bodies are primarily hosted in the coarse-grained dolomite of the Baizuo Group and locally in the lower part of the Zaige Group. This Pb–Zn mining area exhibits a trend of approximately 800 m at an angle of 20–30° NE and a SE-tendency with a big angle of 50–76°. The vertical extension exceeds 1800 m, surpassing its horizontal extension (Fig. 2A). Most ore bodies are bedded, veined, stockwork, and lensed shapes and are distributed along the interbedded faults and fractured zones (Han et al. 2007). The boundary and contact relationships between these Pb–Zn ore bodies and wall rock indicate an epigenetic ore-forming characteristic.

The Kuangshanchang fault extends for approximately 22 km along a 40–65° direction and has a dip angle of 55–73°, which separates the Emeishan basalt from the other underlying strata. This mining area includes 260 unequal-sized Pb–Zn ore bodies with metal reserves of > 3

Mt and 30% Pb + Zn grades. All Pb–Zn ore bodies are found in the coarse-grained dolomite of the Baizuo Group (Fig. 2B). The entire Pb–Zn ore body in the Kuangshanchang mining area exhibits a significant vertical extension of > 1600 m, while its horizontal extension only ranges from 26 to 233 m. From shallow to deep, the Pb–Zn ore body changes from an oxide ore body through a mixed ore body to a sulfide ore body (Fig. 2B; Wang et al. 2016). Each ore body is distributed within the interbedded fault zone in bedded, lentoid, lenticular, and irregular shapes (Zhang et al. 2016).

Three distinct ore-forming periods are reported in the Huize region, represented by sedimentary diagenetic, hydrothermal ore-forming, and supergene periods (Supplementary Fig. 2). The long-term passive continental margin environment since the Cambrian resulted in thick-layered marine carbonate rocks in this area (Supplementary Fig. 1), represented by interbedded dolostone (dolomite) and limestone (calcite) with minor barite, pyrite, and organic matter (e.g., bitumen; Supplementary Fig. 2; Han et al. 2007; Bao et al. 2017; Meng et al. 2019). The hydrothermal ore-forming period comprises three stages: (1) dolomite–pyrite, (2) dolomite–galena–sphalerite, and (3) calcite. Stage I is dominated by abundant fine-grained pyrite (~60 vol.%) and dolomite, associated with negligible calcite, galena, sphalerite, and bitumen (Fig. 2C). Stage II includes galena–sphalerite of > 95 vol% (with sphalerite being three times more abundant than galena) and a small quantity of dolomite, pyrite, and calcite, together with rare microscale arsenopyrite, bitumen, boulangerite, and quartz (Fig. 2C and D). The hydrothermal veins in stage II crosscut those formed during stage I (Fig. 2C and D). Stage III comprises calcite veins without sulfides that either cut the hydrothermal ore veins from stages I and II, or reopen preexisting fractures (Fig. 2D and E). Finally, minor hematite, smithsonite, siderite, and limonite were formed during the supergene period.

Sampling and analytical methods

Sampling

We collected approximately 200 samples from the Qilinchang and Kuangshanchang mining areas, including samples in the underground tunnels (nos. 1031, 1225, 1261, 1272, 1308, 1331, and 1571 sublevels of Qilinchang and nos. 1656, 1740, and 1889 sublevels of Kuangshanchang), drill holes, and field surface mineralization. All samples cover three hydrothermal ore-forming stages in the Huize deposit, and the samples subjected to supergene processes were avoided. Selected specific analyzed regions for each sphalerite color (C1–C6) are shown in Supplementary Fig. 3A–R.

Micro X-ray fluorescence (μ -XRF)

The sphalerite-rich samples were analyzed with a μ -XRF to detect variations in elements such as Zn, Fe, S, and Pb. The analysis was conducted at Guangzhou Tuoyan Testing Technology Co. Ltd, using a Bruker M4 Plus desktop scanning X-ray fluorescence spectrometer equipped with a 20 μ m diameter polycapillary X-ray lens. Two Xflash silicon-drift detectors (SDDs) with a sensitive surface of 60 mm² each were used to improve signal collection and overcome any Bragg peak contamination of the fluorescence spectra (Pagès et al. 2019). The excitation source was based on an Rh X-ray tube operating at 50 kV, 300 μ A, and a vacuum pressure of 2 mbar (Holwell et al. 2016). Sample surfaces were meticulously cleaned of any dust and powder before analysis. Then, each sample was securely affixed onto the motorized stage with metallographic plasticine. The sampling step size used in the mapping analysis was between 20 and 30 μ m, and the dwell time was adjusted to 10–15 ms. Raw data were processed utilizing the M4 Tornado software to obtain element distribution maps. Due to elevated detection limits and potential element interference, most trace element signals (e.g., Ge, Ga, Mn, and Cu) are unavailable through the μ -XRF mapping.

Electron probe micro-analysis (EPMA)

Microprobe measurements were conducted at the State Key Laboratory of Geological Processes and Mineral Resources, China University of Geosciences (Wuhan). The surface of the thin section was uniformly coated with a carbon film (~20 nm thickness), using a JEOL JEE-420 vacuum evaporator (JEOL Ltd., Tokyo, Japan; Zhang and Yang 2016). The analysis was performed using a JEOL JXA-8100 electron probe microanalyzer equipped with four wavelength-dispersive spectrometers. During the analysis, the instrument parameters included an acceleration voltage of 15 kV, an acceleration current of 20 nA, and a beam spot of 2 μ m diameter. The background measurement time was half of the peak measurement time, which remained constant throughout the analysis. The analyzed elements (dwell time) include Zn (10 s), S (10 s), Fe (20 s), Cd (30 s), Cu (30 s), Ge (30 s), Ga (30 s), Ag (30 s), Pb (30 s), and Mn (30 s). The standard samples (corresponding element) are sphalerite (Zn), sphalerite (S), pyrite (Fe), elemental cadmium (Cd), cuprite (Cu), elemental germanium (Ge), gallium arsenide (Ga), and rhodonite (Mn) with detection limits of 0.05%, 0.01%, 0.02%, 0.01%, 0.01%, 0.01%, and 0.01%, respectively. All data were corrected online using a ZAF (atomic number, absorption, and fluorescence) method.

Laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS)

Trace elements analyses were performed using LA-ICP-MS at Nanjing FocuMS Technology Co. Ltd. The instrument was a 193-nm ArF excimer laser system (Analyte Excite) manufactured by the Teledyne CETAC Technologies, which was connected with a quadrupole ICP-MS (Agilent 7700x) manufactured by the Agilent Technologies. Prior to the formal analysis, pre-denudation was performed to eliminate any potential sample-surface contamination. The ultraviolet beam had an energy density of 6.06 J/cm² and a frequency of 6 Hz. A laser beam with a spot diameter of 40 μ m focused on the sample surface for 40 s in order to obtain the elemental composition. The powder sample was mixed with the carrier gases (He and Ar) into the ICP-MS system for analysis. The USGS standard samples MASS-1 and GSE-1G were the combined calibration for sphalerite during the analysis. Off-line selection and integration of background and signals, time-drift correction, and quantitative calibration were performed by ICPMSDataCal software (Liu et al. 2008).

The in situ data obtained from LA-ICP-MS were subjected to principal component analysis (PCA) through a PCA module of the Origin software (e.g., Koch et al. 2012; Frenzel et al. 2016). The elements for PCA included Zn, S, Fe, Mn, Cd, As, Pb, Sn, Ag, Sb, Ge, Ga, and Cu. Among these elements, the Zn and S contents were from EPMA data, and the other elements were derived from corrected LA-ICP-MS data. Before using PCA, we deleted the data below the detection limit, and the remaining data were log-transformed for normality treatment.

The LA-ICP-MS mapping of sphalerite was also conducted at Nanjing FocuMS Technology Co. Ltd, using a 193-nm ArF excimer laser system equipped with an Agilent 7900 quadrupole ICP-MS system. The laser ablation system is furnished with a two-chamber sample compartment designed by the Laurin Technic Company, which offers ample space and rapid purge capabilities. The mapping analysis is performed in a continuous parallel line-scanning mode, where the distance between each line-scan corresponds to the beam spot size. Considering the extensively analyzed area under analysis, we employed a large beam spot (100–120 μ m) with a frequency of 10 Hz and an energy level of 2–3 J/cm². Helium was the carrier gas used during denudation, while argon served as the complementary gas. Before and after the sample analysis, standard samples (NIST 610 or GSE-1G) were measured together, and background signals were collected for 30 s, respectively. The raw data were processed and imaged using the LIMS software program (based on MATLAB), which can automatically complete the signal drift correction and background subtraction (e.g., Xiao et al. 2018).

Electron backscattered diffraction and scanning electron microscope (EBSD-SEM)

The EBSD-SEM measurements were performed at the Geosciences Montpellier laboratory of the University of Montpellier (France) using a Camscan Crystal Probe X500FE scanning electron microscope. Prior to analysis, the sections were further polished and positioned in the sample room with a tilt angle of 70° under a low vacuum of 2 Pa (Demouchy and Tommasi 2021). The analytical conditions employed were an acceleration voltage of 22 kV, a beam current of 5 nA, and a working distance of 25 mm. For the energy dispersive X-ray spectroscopy (EDS)-SEM chemical mapping, different accuracy requirements dictated dwell times ranging from 5 to 15 μ s, pixel sizes between 0.1 and 1.5 μ m, and step sizes spanning from 0.2 to 2.5 μ m (Fernandes et al. 2021). In the EBSD mode, the probe was positioned at a distance of 125 mm (14 mm from the sample), and a Hough resolution of 60 was utilized. The EBSD raw data were processed using the Channel 5 software suite (Oxford Instrument).

Synchrotron X-ray fluorescence (SXRF) mapping and μ -XANES

The synchrotron analyses were conducted on the micro-X-ray absorption spectroscopy (XAS)/micro-XES station of the BM23 beamline at the European Synchrotron Radiation Facility (Grenoble, France; Mathon et al. 2015). A double-crystal monochromator equipped with Si(111) crystals is used to select and scan the energy during acquisitions. An incident X-ray beam was focused down to $\sim 3 \times 3 \mu\text{m}^2$ using focusing mirrors in the Kirkpatrick–Baez geometry. For SXRF mapping, a single-element vortex silicon-drift energy-dispersive detector (SDD) connected to an XIA FalconX1 DSP acquisition electronic was employed to record signals from Fe $K\alpha$ -lines. The Ge signals for SXRF mapping and μ -XANES were measured simultaneously in the high-energy-resolution fluorescence detection (HERFD) mode using a five-crystal-analyzer spectrometer (CAS) in Johann geometry with a 0.5 m bending radius. The HERFD detection used an AXAS-M1 Ketek SDD coupled to an XIA FalconX1 DSP acquisition electronic. The five analyzer crystals were individually tuned to the maximum of the Ge $K\alpha_1$ fluorescence line using a Si(660) Bragg reflection at 78.445°. Using CAS to detect the low-concentrated Ge ($K\alpha_1$ fluorescence line = 9886.42 eV) was motivated by the strong overlap with the highly concentrated Zn ($K\beta_1$ line = 9572 eV), hampering useful XRF and XAS data collection. The energy resolution of the CAS system in this configuration was estimated to be 0.5 eV.

For SXRF mapping, the operating energy was 12.5 keV, with a step size of 10 μ m and a dwell time of 0.5 s. The

raw data were treated using the batch-fitting process of the PyMCA software, and the final elemental mapping was exported (Solé et al. 2007). Four points of interest were selected for high-resolution (HR) μ -XANES analysis, with each spectrum requiring a scan time of 26 min. Three to five spectra were recorded on each point of interest to enhance the signal-to-noise ratio. Subsequently, the spectra were normalized and processed using the Athena software (Ravel and Newville 2005).

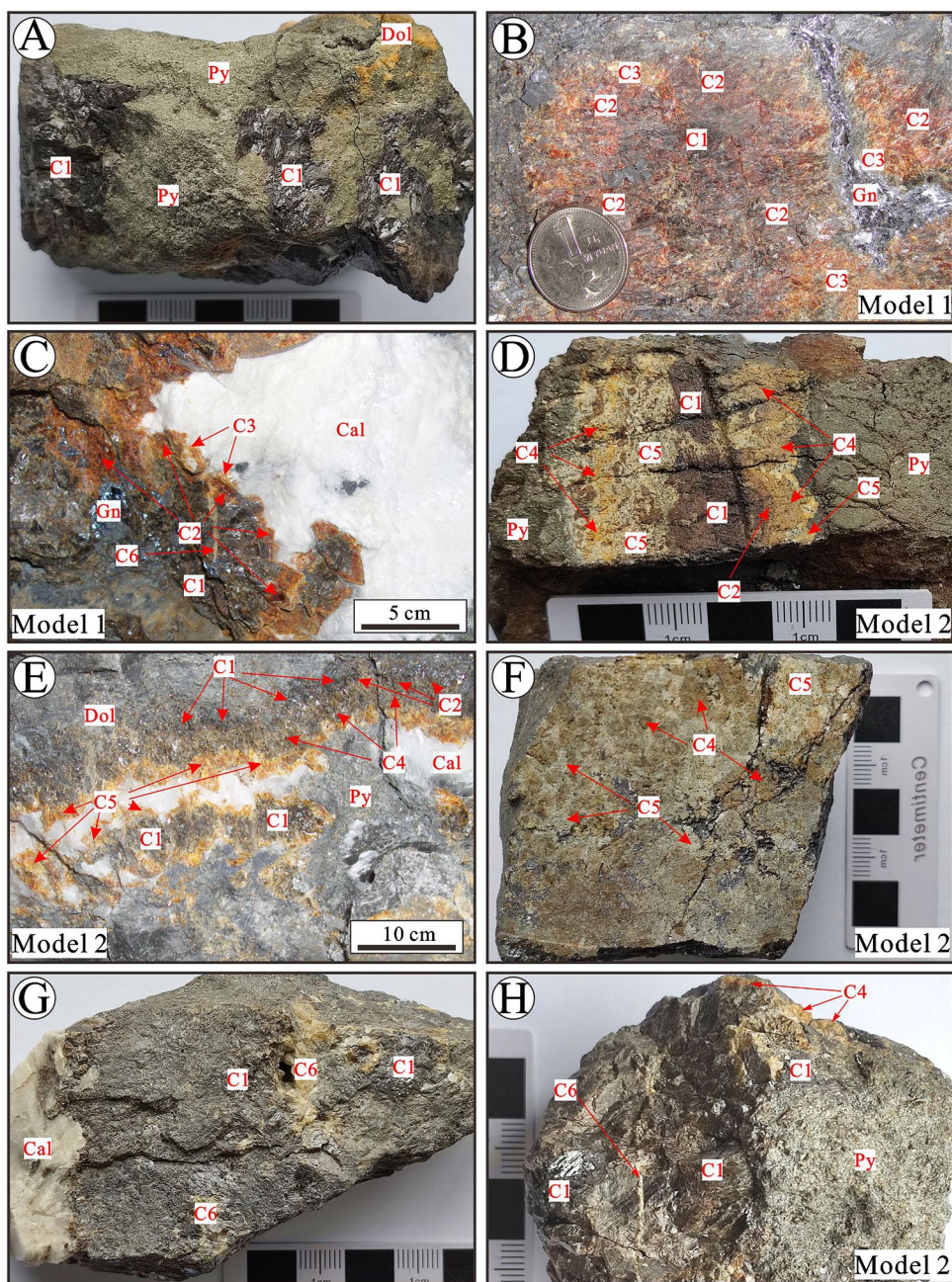
We utilized the crystallographic data of sphalerite (Skinner 1961) to computationally simulate the corresponding theoretical spectrum based on the FEFF10 software (Rehr et al. 2010). In order to investigate the impact of different elemental substitutions, Ge, vacancy, and Cu units were employed as replacements for Zn atoms within the sphalerite structure. The calculations incorporated a full multiple scattering (FMC) range of 5.5 Å and a self-consistent field (SCF) radius of 5.5 Å.

Results

Texture and sphalerite colors

The sulfides in stage II are predominantly composed of sphalerite and galena, with minor amounts of pyrite. Sphalerite mineralization appears in the paragenetic sequence after pyrite and is syn-early galena (Fig. 3A, B; Supplementary Fig. 2). In all the studied samples, sphalerite is commonly found in the vein, massive and replacement texture, exhibiting minimal deformation (Fig. 3A–H). Sphalerite euhedral grains are typically coarse and are filled and enveloped within veins by later-formed sphalerite (Fig. 3C–H). Different sphalerite colors with rhythmic bands were observed in stage II, including black (C1), brown–red (C2), orange–red (C3), deep yellow (C4), pale yellow (C5), and white (C6). Two sphalerite sequence models were identified: Model 1 (C1 $\rightarrow$ C2 $\rightarrow$ C3 $\rightarrow$ C6) and Model 2 (C1 $\rightarrow$ C2 $\rightarrow$ C4 $\rightarrow$ C5 $\rightarrow$ C6). Given its close relationship with pyrite in stage I, the C1 sphalerite was the first to precipitate among all the sphalerite colors (Fig. 3A). In Model 1, the concentric ring structure of C1 $\rightarrow$ C2 $\rightarrow$ C3 sphalerite is crosscut by a composite vein of galena and boulangerite in stage II (Fig. 3B). Another similar concentric ring extends from central C1 through intermediate C2 to C3 sphalerite, which is intersected by C6 sphalerite (Fig. 3C). In Model 2, the C1, C2, C4, and C5 sphalerite can form the rhythmic bands between two pyrite domains in stage I, indicating a forming sequence from C1, through C2 and C4, to C5 sphalerite (Fig. 3D). Between the dolomite–pyrite vein in stage I and the calcite vein in stage III, the C1, C2, C4 and C5 sphalerite colors occur in sequence (Fig. 3E). The spotted distribution of C4 sphalerite is parceled by C5 sphalerite,

Fig. 3 Different colors of sphalerite in the Huize Pb–Zn deposit. **A** The C1 sphalerite closely associated with pyrite in stage I; **B** the concentric ring from C1 through C2 to C3 sphalerite crosscut by the galena and boulangerite composite vein; **C** concentric ring zone of C1 → C2 → C3 → C6 sphalerite; **D** a distinct inside-out zone of C1 → C2 → C4 → C5 between pyrites in stage I; **E** the C1, C2, C4, and C5 sphalerite bands arranged in sequence between pyrite in stage I and calcite in stage III; **F** the spotted C4 parceled by C5 sphalerite; **G** and **H** the C6 sphalerite intersperses and fills with other colors of sphalerite. C1 indicates black, C2 indicates brown-red, C3 indicates orange-red, C4 indicates deep yellow, C5 indicates pale yellow, and C6 indicates white. Gn galena/boulangerite, Py pyrite, Dol dolomite, Cal calcite

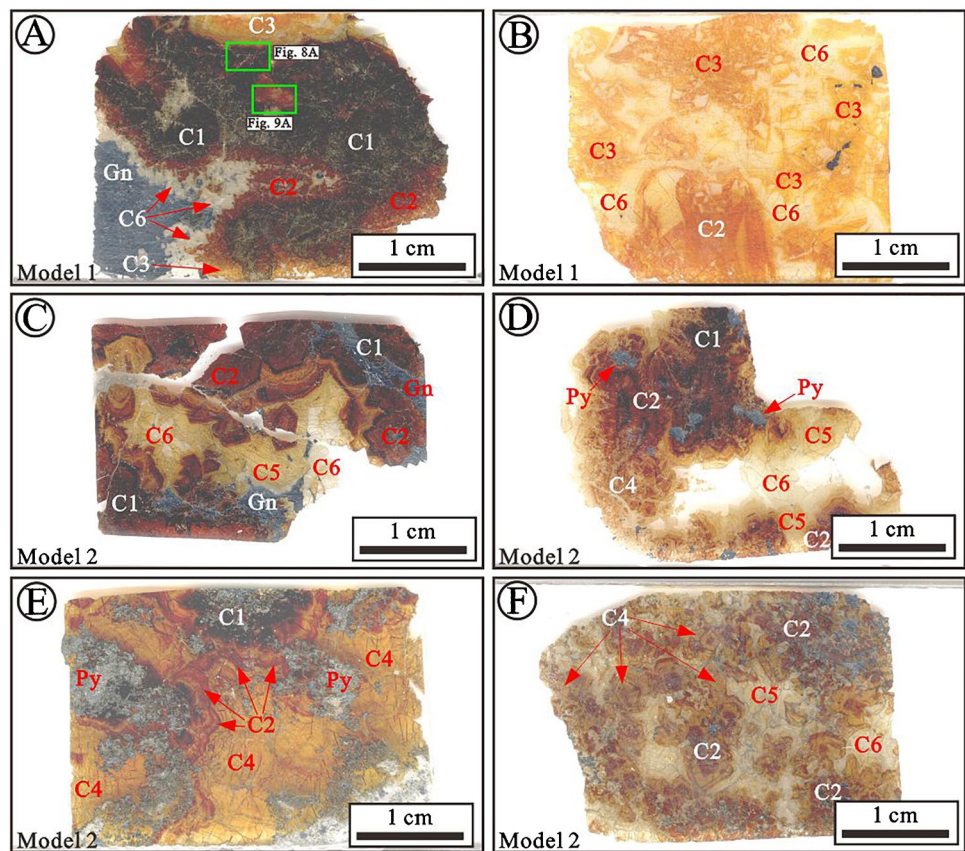


suggesting its earlier formation than that of C5 sphalerite (Fig. 3F). Additionally, the C6 sphalerite cuts through and fills in other colors of sphalerite grains such as C1 and C4 (Fig. 3G and H).

In Model 1, the massive C1 sphalerite is early compared to C2, C3, and C6 sphalerite bands as well as galena, indicating a sequential formation of C1 → C2 → C3 → C6 → galena (Fig. 4A). The C6 sphalerite grains typically enclose the C2 and C3 sphalerite and exhibit a close association with galena in stage II (Fig. 4A

and B). In Model 2, the C1 and C2 sphalerite domains are subsequently replaced by and filled with the C4, C5, and C6 sphalerite veins (Fig. 4C and D). The interlaced band structure of C1, C2, and C4 sphalerite assemblages can be observed in the C4 and C5 sphalerite (Fig. 4E and F). Among these sphalerite colors, the C3 sphalerite in Model 1 and C4 and C5 sphalerite in Model 2 do not overlap within the same sample, confirming that the two sphalerite sequence models seem to exist independently without any distinct symbiosis relationship (Figs. 3A–H and 4A–F).

Fig. 4 Representative photos of sphalerite colors in the Huize Pb–Zn deposit. **A** The mineral assemblages of C1 → C2 → C3 → C6 → galena; **B** the C2 and C3 sphalerite were replaced by C6 sphalerite; **C** the C1 and C2 sphalerite showed a rhythmic structure and were replaced by C5 and C6 sphalerite; **D** and **E** pyrite, C1 and C2 sphalerite were replaced by C4, C5 and C6 sphalerite; **F** the C1 and C4 sphalerite showed a rhythmic structure and were replaced by C5 and C6 sphalerite. C1 indicates black, C2 indicates brown-red, C3 indicates orange-red, C4 indicates deep yellow, C5 indicates pale yellow, and C6 indicates white. Gn galena/ boulangerite, Py pyrite



μ-XRF mapping

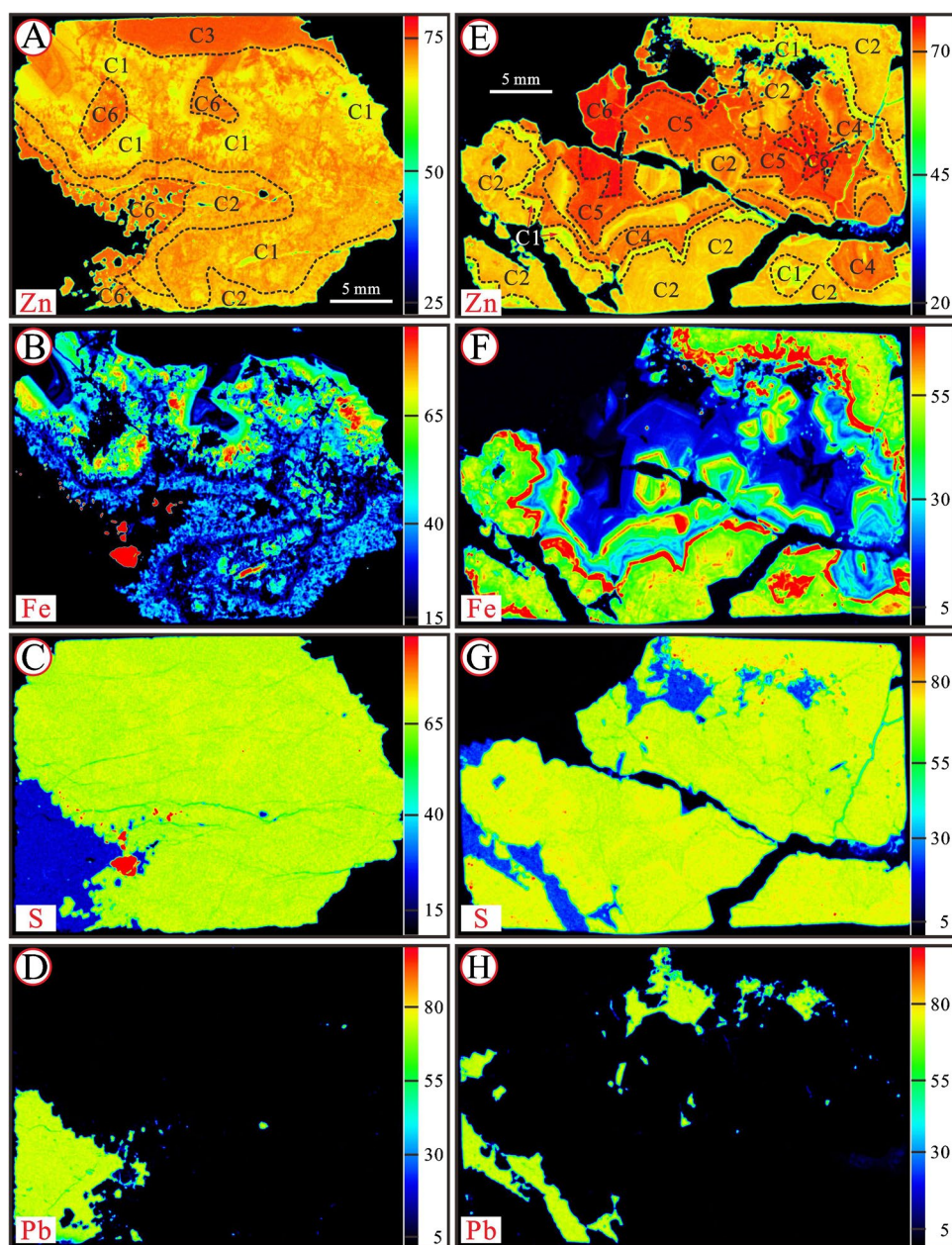
Two sphalerite samples (two models) with rhythmic bands were analyzed using the μ-XRF (Fig. 4A and C), and the reported elements include Zn, S, Fe, and Pb. Sample 1 (KS-1740–25-2) comprises the internal C1 sphalerite and peripheral C2, C3, and C6 sphalerite bands with minor galena (Fig. 4A). There is a small amount of C6 sphalerite distributed within the C1 sphalerite. In the element mapping, the C3 and C6 sphalerite grains show the most vital Zn signals but the weakest Fe signals (Fig. 5A and B). From the C1 to C2 sphalerite, the element signals progressively improve for Zn but decrease for Fe. However, most of the C1 and C2 sphalerites have intermediate characteristics without significant differences in Zn and Fe signals (Fig. 5A and B). The above four sphalerite colors exhibit similar S but weak Pb signals, with the Pb signal primarily attributed to galena (Fig. 5C and D). Sample 2 (QL-1331–4-2) comprises the C1, C2, C4, C5, and C6 sphalerite bands, displaying a distinct rhythmic zone along with minor galena (Fig. 4C). From the C1 and C2 to C4, C5, and C6 sphalerite grains, a corresponding decrease in Fe signal intensity accompanies a gradual increase in Zn signal intensity. This transition is highlighted through color changes ranging from green–yellow–orange to light–deep red for Zn and red–green–blue

to deep blue for Fe (Fig. 5E and F). Notably, no Fe signal is observed in the element mapping of the C6 sphalerite (Fig. 5F). The C1 sphalerite has the highest S signal values (red), of which C2, C4, C5, and C6 are somewhere between yellow and green (Fig. 5G). The Pb signals in this sample result from prominent galena (Fig. 5H).

EPMA spot analysis data

In total, 360 selected points (60 per colored type) were analyzed for their major element compositions (Supplementary Table 1), and the box diagrams of Zn, S, and Fe contents are presented in Fig. 6. The Zn contents in sphalerite exhibit a gradual increase from C1 (54.84–65.35%, median 59.38%) and C2 (59.57–62.14%, median 60.73%), to C4 (63.49–65.33%, median 64.41%), C5 (63.94–65.68%, median 64.70%), and C6 (65.51–67.40%, median 66.30%). In contrast, the Fe content of sphalerite gradually decreases from C1 (1.33–10.78%, median 6.90%) and C2 (4.10–6.34%, median 5.39%), to C4 (1.37–2.35%, median 1.86%), C5 (1.23–2.13%, median 1.72%), and C6 (0.01–0.27%, median 0.05%). The Zn (64.55–67.23%, median 66.34%) and Fe (0–1.41%, median 0.03%) contents in C3 sphalerite are comparable to those of C6 sphalerite (Fig. 6). The S content of sphalerite exhibits an overall decreasing

Fig. 5 The μ -XRF images of two representative sphalerite samples, showing single Zn, Fe, S, and Pb signals, including **A–D** sample 1 (KS-1740-25-2) and **E–H** sample 2 (QL-1331-4-2). C1 indicates black, C2 indicates brown-red, C3 indicates orange-red, C4 indicates deep yellow, C5 indicates pale yellow, and C6 indicates white



trend from C1 (32.90–33.96%, median 33.50%), through C2 (32.75–33.87%, median 33.36%), C3 (32.68–33.47%, median 33.21%), C4 (32.82–33.57%, median 33.20%), C5 (32.75–33.63%, median 33.20%), to C6 (32.62–33.62%, median 33.10%). Supplementary Table 1 shows the analyzed trace elements of Cd, Ga, Ge, Mn, and Cu.

LA-ICP-MS mapping

The LA-ICP-MS mapping analysis was performed on a sphalerite domain with distinct colors, as depicted in Fig. 7A. Four sphalerite colors are observed in this region, complying with a transition band of C1 → C2 → C5 → C6

from left to right (Fig. 7B). Only the signal changes of Fe and Mn correspond to the color bands of sphalerite (Fig. 7C–N). No apparent Zn and S changes are observed across all colored sphalerite domains, except for weak signals originating from galena located in the lower right corner (Fig. 7C and D). In contrast, Fe, Cu, Mn, Ag, Cd, Ga, Ge, and Sn show significant signal changes (Fig. 7E–K and M). Along the C1 → C2 → C5 → C6 sphalerite sequence, both Fe and Mn signals decrease gradually (Fig. 7E and G). There are two modes of element distribution: (1) Cu, Ag, Ge, and Sn exhibit enrichment exclusively in C1 sphalerite; (2) Ga and In are depleted in C1 sphalerite but enriched in other sphalerite colors (Fig. 7F, H, K, and M). All sphalerite

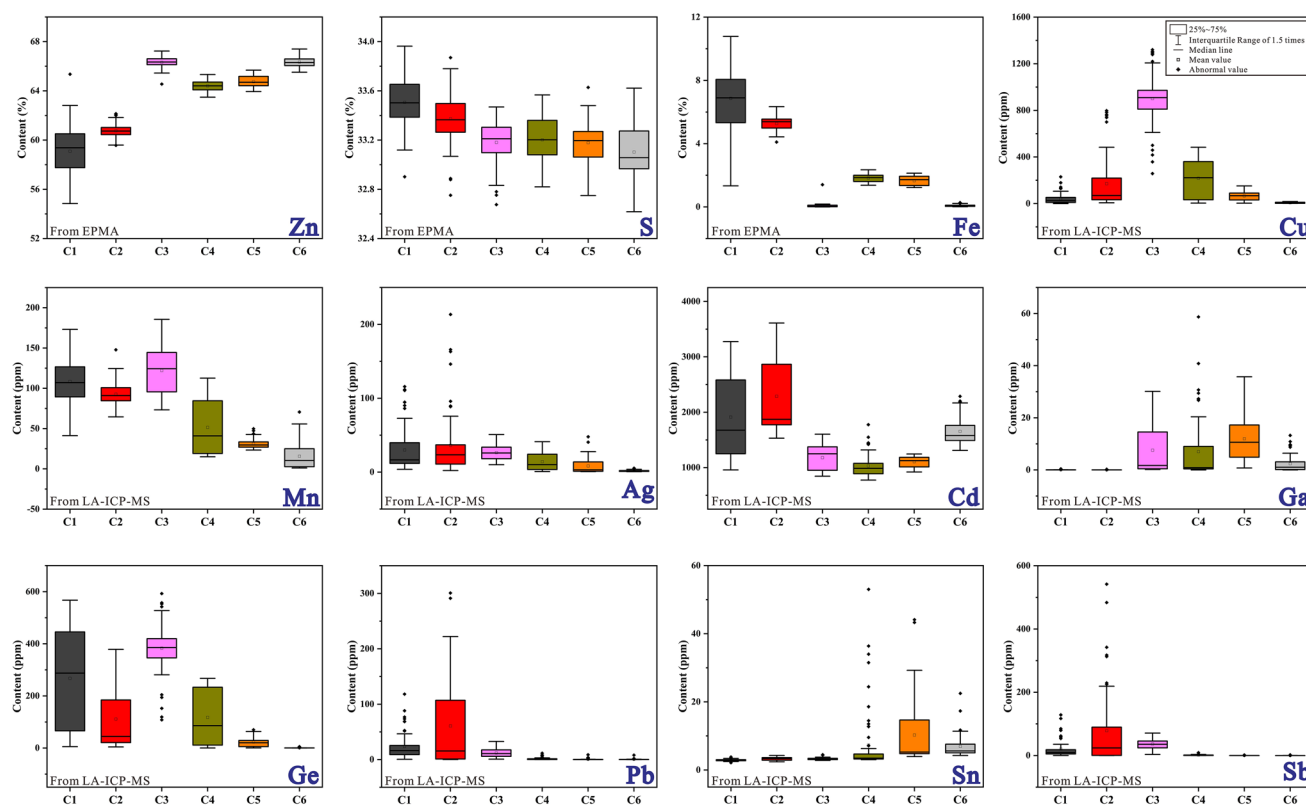


Fig. 6 The box-and-whisker diagrams of analyzed element contents (Zn, S, Fe, Cu, Mn, Ag, Cd, Ga, Ge, Pb, Sn, and Sb) in sphalerite from the Huize Pb–Zn deposit

colors display strong Cd (red) and weak Sb (blue) elemental signals without regular distribution (Fig. 7I and N). A small amount of galena at the lower right contributes to the primary element signals of Ag and Sb (Fig. 7H and N).

LA-ICP-MS spot analysis data

In total, 357 LA-ICP-MS in-situ data (excluding three data points affected by microscale mineral inclusions) corresponding to analyzed EPMA points are used to represent the trace element compositions of six sphalerite colors, and Supplementary Table 2 summarizes the results. Most trace elements exhibit smooth curves on the time-resolved LA-ICP-MS signals (Supplementary Fig. 4A–F). However, certain elements display signal profiles with peaks in specific samples, for example, Pb and Sb in the C1 sphalerite (Supplementary Fig. 4G), Na, K, and Sr in the C3 sphalerite (Supplementary Fig. 4H), Mg and Al in the C5 sphalerite (Supplementary Fig. 4I), and Si in the C6 sphalerite (Supplementary Fig. 4J). These abnormal signal curves do not lead to cooperative changes of other analyzed elements (Fe, Cu, Mn, Ag, Cd, Ga, Ge, In, and Sn), indicating the limited influence from minor mineral inclusions observed at the microscale within sphalerite.

The C3 sphalerite hosts the highest Cu content (258–1321 ppm, median 910 ppm), followed by the C2 of 7–798 ppm and C4 of 4–483 ppm (Fig. 6). However, the C1, C5, and C6 sphalerite data show significantly low Cu contents, especially the C6 sphalerite (1–18 ppm).

The Mn is enriched in the C1, C2, and C3 sphalerite, with contents of 41–173 ppm (median 107 ppm), 65–148 ppm (median 91 ppm), and 73–186 ppm (median 124 ppm), respectively. These values are slightly higher than the Mn contents found in the C4 (15–113 ppm), C5 (23–50 ppm), and C6 sphalerite (1–71 ppm; Fig. 6).

The Cd concentrations in the C1 and C2 sphalerite are 958–3275 ppm and 1531–3610 ppm, respectively (Fig. 6). The C6 sphalerite also contains an apparent Cd component between 1310 and 2288 ppm, albeit slightly lower than that in C1 and C2 sphalerite. Conversely, the Cd contents of C3, C4, and C5 sphalerite are significantly diminished (Fig. 6).

The C1 and C3 refer to the Ge-rich sphalerite, with most Ge contents in the C3 sphalerite exceeding 300 ppm, while the Ge contents in the C1 sphalerite range from 6 to 567 ppm (Fig. 6). The C2 and C4 sphalerite colors show Ge concentrations of up to 379 ppm, much higher than the Ge contents in the C6 sphalerite (<5 ppm; Fig. 6).

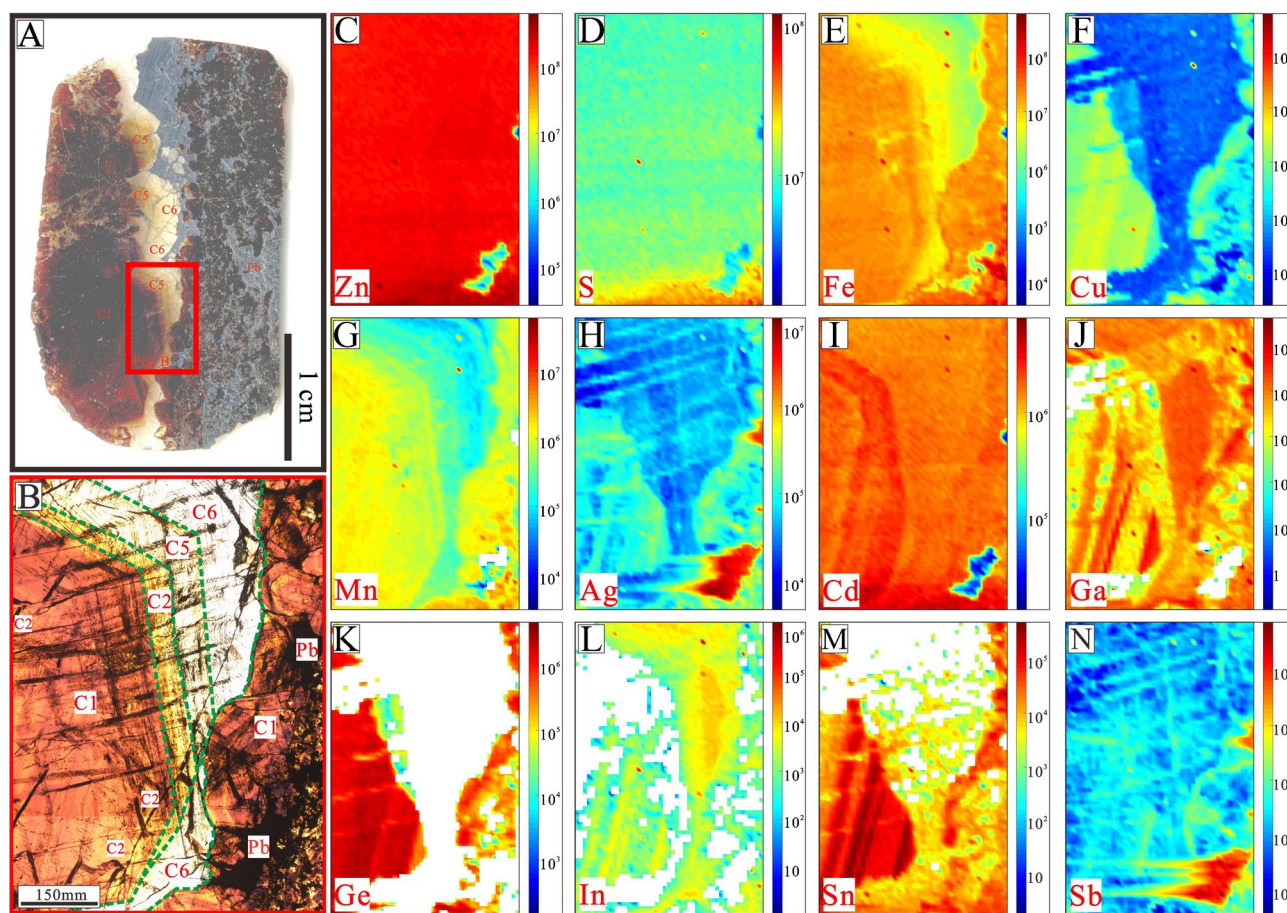


Fig. 7 Laser ablation (LA) inductively coupled plasma (ICP) mass spectroscopy (MS) element mapping of sphalerite sample QL-1225-8 in the Huize Pb–Zn deposit. **A** and **B** the transmitted light photograph

of the selected sample and analyzed region; **C–N** element signals of Zn, S, Fe, Cu, Mn, Ag, Cd, Ga, Ge, In, Sn, and Sb

Only the C2 sphalerite contains a certain amount of Pb element (0–301 ppm, median 16 ppm), followed by the C1 and C3 sphalerite (Fig. 6). The Pb levels in C4, C5, and C6 sphalerite are negligible. The contents of other trace elements are provided in Supplementary Table 2 and presented in Fig. 6.

PCA result and correlation matrix of sphalerite composition

Thirteen principal components (PC) represent 100% of the information from 357 data in the studied sphalerite, where PC1 and PC2 display the primary information. Supplementary Fig. 5A–L shows the corresponding results, including the eigenvalues, percentage of variance, cumulative principal component number (nos. 1–13), loading values, and binary diagrams of PC1 and PC2. There is a strong correlation between Ge and Cu contents in the C3 sphalerite, followed by the C2, C4, and C5 sphalerite (Supplementary Fig. 5D, F, H, J). However, no apparent

correlation appears between Ge and Cu in the C1 and C6 sphalerite (Supplementary Fig. 5B, L). Except for Cu, no element shows a noticeable correlation with Ge in all sphalerite colors. The PCA was also conducted on all data from six sphalerite colors, and PC1 (42.4%) and PC2 (25.5%) are used to represent the overall information (Supplementary Fig. 6). Among these sphalerite colors, C1 and C2 exhibit similar characteristics, but with a wider range of data observed for C1. On the other hand, the data points for C3 sphalerite are highly concentrated and represent a distinct characteristic compared to other colors. From C4, through C5, to C6 sphalerite, there is a gradual decrease in their data ranges, with a trend toward negative values of PC1. The contents of Mn, Fe, Cu, Ga, Ge, As, Ag, Cd, In, Sn, Sb, Pb, and Zn from different sphalerite colors were used to perform the correlation matrix (Supplementary Table 3). The C3 sphalerite also shows the strongest correlations between Ge and Cu (correlation index, 0.99), followed by the C4 (0.94), C5 (0.94), and C2 (0.82) sphalerite colors. In addition, only Ag displays some weak

correlations with Ge contents, for example, the C4 (0.74) and C3 (0.59) sphalerite (Supplementary Table 3).

EBSD data

Figure 8A represents the backscattered electron (BSE) image of the selected region for mapping analysis, where Ge signals are marked with yellow circles of varying scales to visualize their distribution better. Most Ge signals are distributed inside the single sphalerite grains in the Euler color image and inverse pole figures (Supplementary Fig. 7A–D). A typical Ge-bearing mineral with a particle size of ~20 µm was selected for the frame-shift-wake scanning (Fig. 8B). It exhibits a dark gray color under the transmitted light, displaying a regular shape and clear boundaries (Fig. 8C). The Ge-bearing mineral is enriched in Ge and O but depleted in Zn and S signals in the element mappings (Fig. 8D–G), further confirmed as argutite (GeO_2) by the XRF spectrum and structure information (Fig. 8H–J).

Synchrotron X-ray fluorescence and μ -XANES spectroscopy

The selected region from Fig. 4A was utilized for SXRF mapping analysis, and the transmitted and reflected light photographs exhibit the C1, C2, C3, and C4 sphalerite domains and analyzed spots for the μ -XANES (Fig. 9A and B). The element mappings of Ge (high resolution) and Fe, along with their local magnifications, are presented in Fig. 9C–F, and the regions with some argutite grains in the Ge mapping are magnified twice for better presentation. The Fe and Ge signals are uneven in the C1 sphalerite, where some argutite grains are observed explicitly in the Ge- and Fe-rich zones (Fig. 9C–F). The C3 sphalerite shows a less contrasted Ge distribution and a more homogenous Fe distribution than the C1 sphalerite. The C2 sphalerite shows the transition characteristics of Fe between C1 and C3 sphalerite but has the lowest Ge signal strength (Fig. 9C and D).

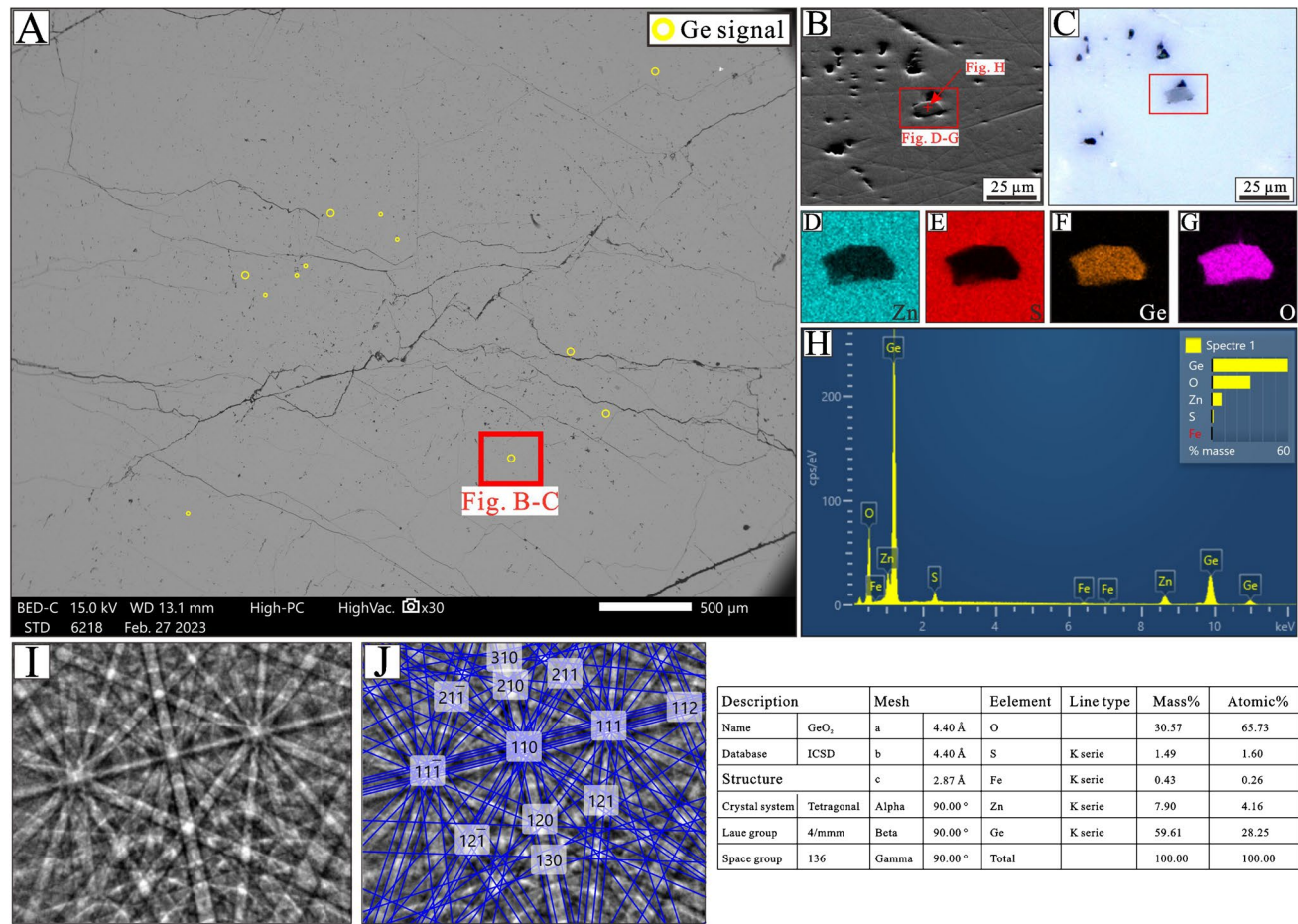


Fig. 8 The electron backscattered diffraction (EBSD) photographs and spectra of sphalerite in the Huize Pb–Zn deposit. **A** The scanning electron microscope (SEM) photo of sphalerite, showing the Ge signals (GeO_2); **B–G** the SEM image and transmitted light photograph

of a representative argutite grain (GeO_2), as well as elemental mappings of Zn, S, Ge, and O; **H** the energy spectrum signal of argutite (GeO_2); **I** and **J** crystal structure information of argutite (GeO_2) based on the EBSD analysis (from a selected point in panel B)

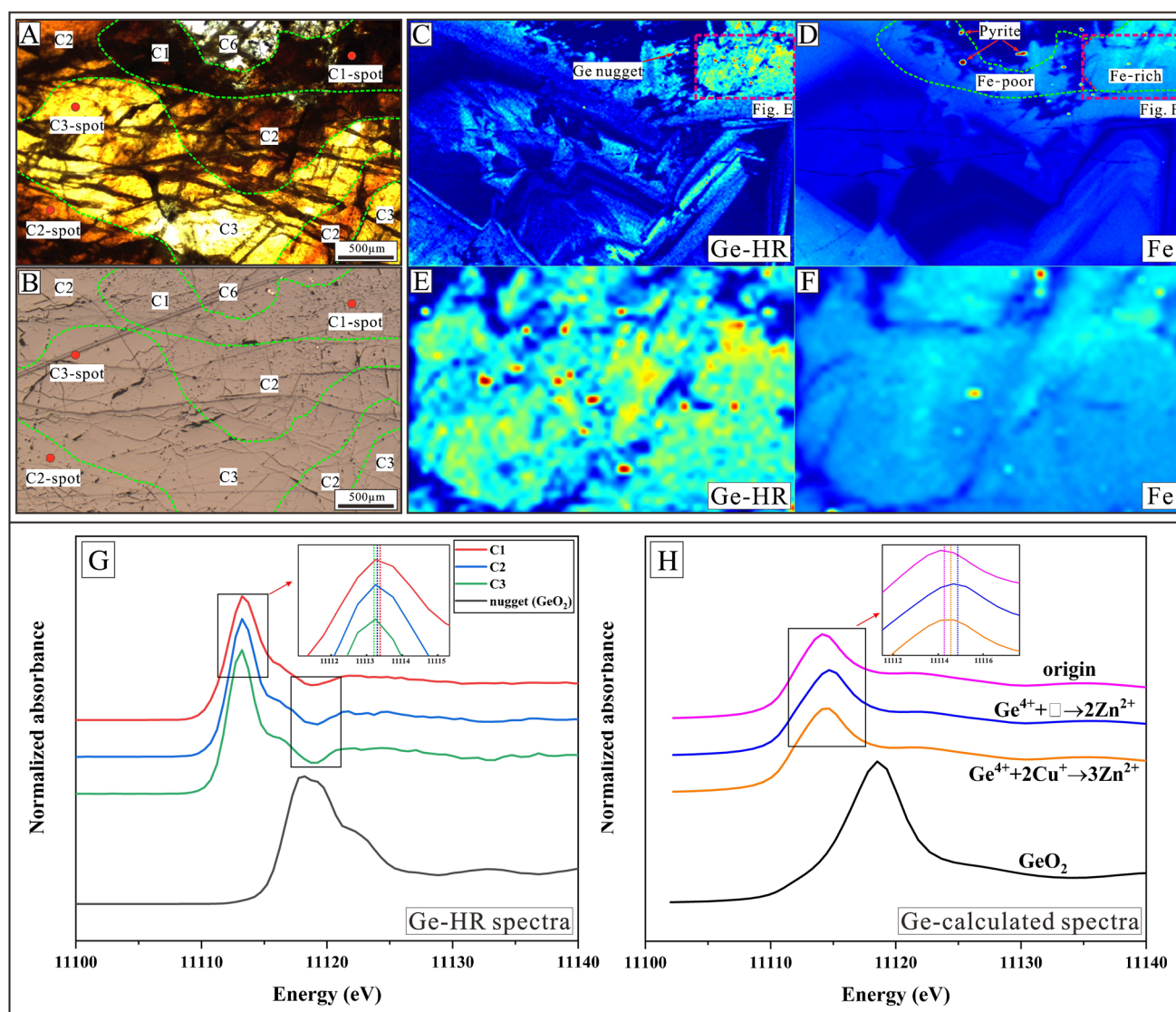


Fig. 9 **A, B** Transmitted and reflected light photographs of analyzed area (from Fig. 4A) within different sphalerite colors and analyzed points; **C–F** the synchrotron X-ray fluorescence (SXRF) maps of Ge-HR and Fe in the selected region, along with their enlarged layout images; **G** and **H** the normalized Ge high-resolution synchrotron

X-ray absorption near-edge structure spectroscopy (μ -XANES) spectra of C1, C2, C3, and nugget in the Huize deposit and corresponding ab initio calculations of the origin, $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 2\text{Zn}^{2+}$, and $\text{Ge}^{4+} + \square \rightarrow 2\text{Zn}^{2+}$ based on sphalerite structure and argutite (GeO_2)

Because of their low Ge concentration, the HR (high-resolution) Ge K-edge μ -XANES spectra for the C4, C5, and C6 sphalerite show a signal-to-noise ratio too low to be exploited. Therefore, only the C1, C2, and C3 were subjected to high-resolution μ -XANES. A small shift towards low energies can be observed at the white line of the XANES spectra (i.e., the maximum observed high after the edge) from C1 through C2 to C3 sphalerite (i.e., 11113.5 eV, 11113.4 eV, and 11113.2 eV, respectively; Fig. 9G). These spectra all exhibit a minimum oscillation at approximately 11118.9 eV, which is progressively accentuated for C1, C2, and C3, respectively. Note that due to the oxygen ligands,

the spectrum obtained for argutite grain displays a white-line position of 11118.1 eV (Fig. 9G).

Considering the strong linear relationship between Ge and Cu in Cu-rich (e.g., C3) sphalerite and the heterogeneous distribution of Ge in Fe-rich (e.g., C1 and C2) sphalerite, we used the FEFF10 software to simulate the XANES spectra for two types of Ge substitutions in sphalerite: (1) $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 3\text{Zn}^{2+}$ and (2) $\text{Ge}^{4+} + \square \rightarrow 2\text{Zn}^{2+}$ (Belissont et al. 2014; Zhuang et al. 2019). We replace the central Zn in the sphalerite structure with one Ge as “origin” (Supplementary Fig. 8A). On this basis, the first-order adjacent Zn atoms are substituted by either two Cu atoms or one vacancy as

“ $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 2\text{Zn}^{2+}$ ” and “ $\text{Ge}^{4+} + \square \rightarrow 2\text{Zn}^{2+}$ ”, respectively (Supplementary Fig. 8B, C). The calculated spectra show minor shifts of Ge edge positions for these three models (11114.8 eV, 11114.6 eV, and 11114.3 eV; Fig. 9H). Therefore, the calculation based on the FEFF10 software concluded that different element substitutions with Ge would only cause slight shifts in the Ge edge position without significant alterations in spectrum shape.

Discussion

Ge-bearing phases in the Huize Pb–Zn deposit

Ge-bearing sphalerite

The LA-ICP-MS analysis used in this study is limited to the microscopic scale. However, some Ge-bearing mineral (e.g., argutite in this study) grains might be evenly distributed in sphalerite as nano-mineral inclusions, as reported by Fougere et al. (2023). In this study, we could not distinguish the Ge as nano-mineral inclusions in sphalerite from the Ge in the sphalerite crystal lattice. The atom probe tomography (APT) and the extended X-ray absorption fine structure (μ -EXAFS) analyses represent potential methods for addressing these challenges. Numerous published LA-ICP-MS data on Huize sphalerite reveal a wide range of Ge concentrations, spanning from 0 to 930 ppm (e.g., Ye et al. 2011; Wu et al. 2019; Tao et al. 2019; Ji 2019; Oyebamiji et al. 2020; Liu et al. 2022; Supplementary Table 4). Most C3 sphalerite compositions show Ge concentrations exceeding 300 ppm, while the Ge content of C1 sphalerite ranges from 6 to 567 ppm (Fig. 6). The C2 and C4 sphalerite colors display notable Ge concentrations between 0 and 379 ppm, which is significantly higher than the Ge content observed in the C6 sphalerite (< 5 ppm; Fig. 6). In this study, the Ge contents in different colored sphalerite are concentrated, except for the C1 sphalerite (Fig. 6). In the C1 sphalerite, the Fe and Ge signals and contents are highly variable; for example, the Fe content ranges from 1.33 to 10.78% (median 6.90%), while the Ge content varies between 6 and 567 ppm (median 288 ppm) (Fig. 6). Additionally, Ge seems to be enriched in Fe-rich region (i.e., top right corner) and depleted in Fe-poor region in the SXRF mappings (Fig. 9C and D). It should be noted that Fe content in sphalerite can be interpreted as controlled by temperature in many submarine volcanic-hosted massive sulfide (VHMS) deposits (Keith et al. 2014). Factors controlling Fe contents such as temperature may therefore be a potential control for higher Ge contents in sphalerite (Frenzel et al. 2016; Liu et al. 2020).

In Model 1 with $\text{C1} \rightarrow \text{C2} \rightarrow \text{C3} \rightarrow \text{C6}$ sphalerite, Ge underwent two enrichment stages (C1 and C3; Fig. 6). The C1 sphalerite represents Fe-rich sphalerite, and the

C3 sphalerite hosts the highest Cu and Mn concentrations (Fig. 6). In contrast, the Ge content decreases gradually in Model 2 of $\text{C1} \rightarrow \text{C2} \rightarrow \text{C4} \rightarrow \text{C5} \rightarrow \text{C6}$ (Fig. 6). Overall, the different sphalerite colors in the two models indicate a gradual decrease in Fe contents, suggesting the decreasing temperatures (Figs. 4A–F and 10A; Keith et al. 2014; Gagnevin et al. 2014). However, the reasons behind the emergence of these two distinct models remain unclear.

According to the published data, the Pb–Zn deposits in the SYG area typically contain Ge contents up to several hundred ppm in sphalerite; for example, the Fule (up to 941 ppm), Lehong (up to 536 ppm), and Maoping (up to 514 ppm) deposits (Ren et al. 2019; Wei et al. 2019; Niu et al. 2023; Supplementary Table 4). Comparatively, other renowned MVT deposits worldwide, including the Lisheen, Alps, and the MVT deposits in the USA and South America, contain Ge up to 200–4000 ppm in sphalerite (Marsh et al. 2016 and references therein). In comparison, the Ge content (up to 593 ppm) in the Huize sphalerite has no apparent advantage, and the significant economic Ge resources of this deposit are attributed to the presence of high-grade Zn ore (> 15%).

Ge-dioxide

Except for the Ge-rich sphalerite (C1 and C3), argutite (GeO_2) with 70 wt% Ge is a newly discovered Ge-bearing phase in this deposit (Fig. 8A). Argutite only occurs in the C1 sphalerite throughout two sequence sphalerite models (Fig. 4A and C). No occurrences of argutite have been identified in the other colors of sphalerite (C2, C3, C4, C5, and C6), even in the C3 sphalerite with the highest Ge contents (median, 386 ppm; Fig. 6). Metamorphism is a frequent explanation for the occurrence of highly enriched Ge-minerals associated with deformed sphalerite, for example, brunogeierite (GeFe_2O_4) and carboirite [$\text{GeFeAl}_2\text{O}_5(\text{OH})_2$] in the Pb–Zn deposits of the Pyrenean Axial Zone, Western Europe (Cugerone et al. 2019). The Ge-bearing independent minerals are typically enriched in late fine sphalerite grains or at the edge of original coarse-grained sphalerite (Cugerone et al. 2019, 2021; Bauer et al. 2019; White et al. 2022). In the Huize deposit, these argutite grains were precipitated simultaneously with C1 sphalerite and have not undergone recrystallization or redistribution. First, the C1 sphalerite is predominantly coarse-grained and has experienced minimal deformation and metasomatism (Figs. 4A and 8A). Second, argutite grains are distributed randomly in the C1 sphalerite rather than along the fractures in the Euler and inverse pole figure (IPF) maps (Supplementary Fig. 7A–D). Therefore, argutite could be formed during the early stage of sphalerite precipitation, when the fluid is still highly enriched in Ge (Han et al. 2007).

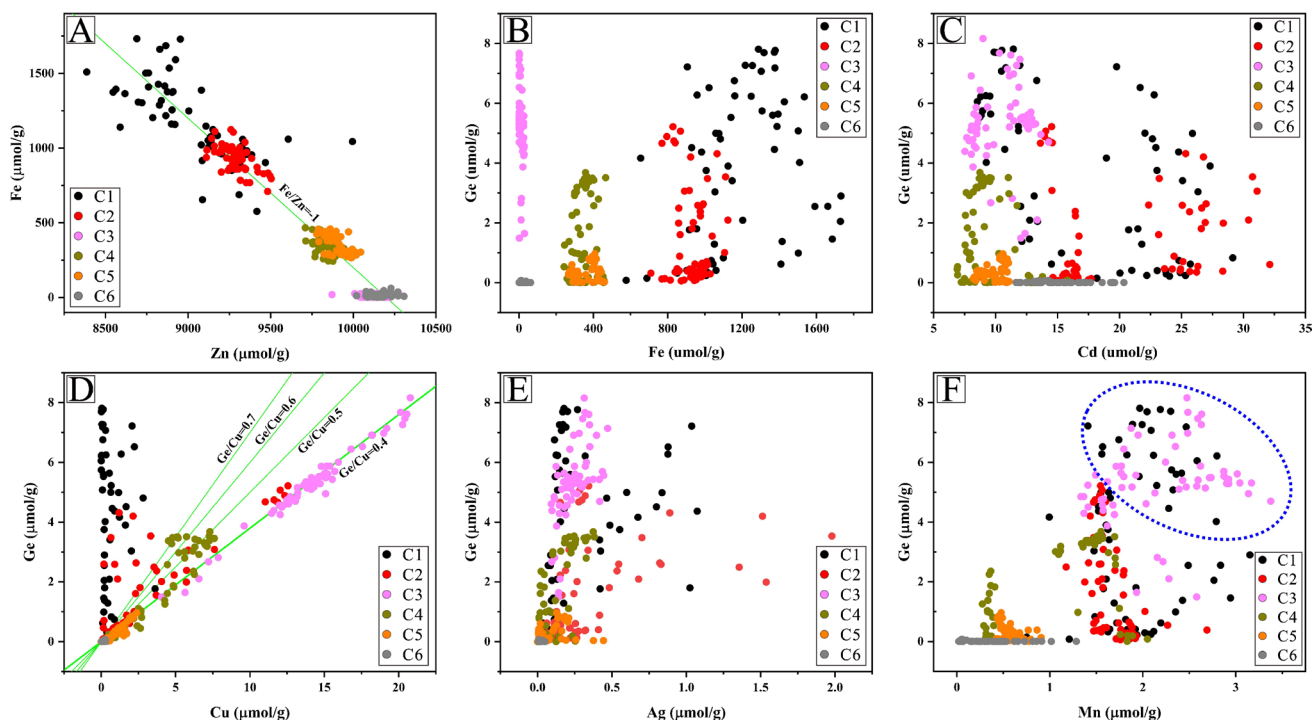


Fig. 10 Binary diagrams of **A** Zn versus Fe, **B** Fe versus Ge, **C** Cd versus Ge, **D** Cu versus Ge, **E** Ag versus Ge, and **F** Mn versus Ge of different colored sphalerite in the Huize deposit

Even though this SYG area is China's most significant economic Ge-producing area, there have been no previous reports on Ge-bearing minerals in the entire SYG Pb–Zn metallogenic province, except for one report on suspected kieselite [$\text{Al}_2(\text{GeO}_4)\text{F}_2$] in the Huize deposit (Zhang et al. 2008). Despite being a newly reported Ge-bearing mineral in this study, argutite is found to be scarce and present as tiny microscale particles (not exceeding 20 μm ; Fig. 8A and B). Based on microscale measurements, our observation indicates that the primary source of Ge in the Huize deposit is within the sphalerite crystal lattice.

Incorporation mechanism of Ge into sphalerite

Fe^{2+} is the predominant element that replaces Zn^{2+} in the sphalerite crystal lattice, which is responsible for the various sphalerite colors in this study, except for the C3 and C6 sphalerite that may be influenced by different Cu content (Figs. 5A, B, E, and F; 6; 7C and E; and 10A; Wright and Gale 2010; Wei et al. 2021). The abundance of Mn and Ga strongly correlates with the Fe signal (Fig. 7G, I, and J). However, we also demonstrate that the Cu signal tightly controls the level of Ge, which also slightly contributes to some other trace elements in the LA-ICP-MS mappings (e.g., Ag, Sn; Fig. 7F, H, K, and M). Fe, Cd, Mn, Cu, Ag, and vacancy ($\square$) can be coupled with Ge into sphalerite, represented by (1) $2\text{Fe}^{2+} + \text{Ge}^{4+} + \square \leftrightarrow 4\text{Zn}^{2+}$, (2) $\text{Ge}^{4+} + 2\text{Ag}^+ \leftrightarrow 3\text{Zn}^{2+}$,

(3) $\text{Ge}^{4+} + 2\text{Cu}^+ \leftrightarrow 3\text{Zn}^{2+}$, (4) $\text{Ge}^{4+} + \square \leftrightarrow 2\text{Zn}^{2+}$, and (5) $\text{Ge}^{4+} + \text{Mn}^{2+} + \square \leftrightarrow 3\text{Zn}^{2+}$ (Cook et al. 2009, 2015; Belissont et al. 2014; Hu et al. 2021). Among these trace element compositions, only Cu strongly correlates with Ge, particularly in the C3, C4, and C5 sphalerite (Fig. 10B–F; Supplementary Fig. 5F, H, J). Numerous synchrotron-based experiments have confirmed that the substitution of Ge^{4+} (excluding Ge^{2+}) for Zn^{2+} occurs in the sphalerite (e.g., Cook et al. 2015; Belissont et al. 2016; Liu et al. 2023). Therefore, Ge in the studied sphalerite (C1–C6) crystal lattice is also considered to be Ge^{4+} .

Combined with the distinct correlation of Ge and Cu, the substitution of $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 3\text{Zn}^{2+}$ is widely present in the C3 sphalerite, while the Ge/Cu ratios approaching 0.4 other than 0.5 can be attributed to the requirement of Cu to balance other trivalent or tetravalent metal ions (e.g., In^{3+} , Ga^{3+} , Sb^{3+} , and Sn^{4+} ; Fig. 6; Murakami and Ishihara 2013; Bauer et al. 2019; Cugerone et al. 2021; Yu et al. 2021). When no element is distinctly related to the Ge content, the vacancy ($\square$) was considered in the Ge substitution (e.g., Cook et al. 2015; Hu et al. 2021). Hence, the presence of Ge in the C1 and part of C2 sphalerite is attributed to the substitution of $\text{Ge}^{4+} + \square_{(\text{vacancy})} \rightarrow 2\text{Zn}^{2+}$, and the introduction of $\text{Fe}^{2+} \rightarrow \text{Zn}^{2+}$ resulted in abundant vacancies that facilitate Ge incorporation into sphalerite (Fig. 10D; Lepetit et al. 2003; Pfaff et al. 2011). The correlation of Cu and Ge contents in the C2, C4, and C5 sphalerite is weaker than that observed in the C3 sphalerite (Supplementary Fig. 5D, H, J;

Supplementary Table 3). Therefore, the Ge substitutions in the C2, C4, and C5 sphalerite could share transitional characteristics between the C1 and C3 sphalerite—a combination substitution of $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 3\text{Zn}^{2+}$ and $\text{Ge}^{4+} + \square_{(\text{vacancy})} \rightarrow 2\text{Zn}^{2+}$ (Fig. 10A and D). Among Fe, Cd, Ag, and Mn, there are some weak correlations of Ag and Ge in the C2, C3, and C4 sphalerite, indicating that the substitution of $\text{Ge}^{4+} + 2\text{Ag}^+ \rightarrow 3\text{Zn}^{2+}$ could potentially occur but locally in this study (Fig. 10B–F; Supplementary Table 3).

From C1, through C2, to C3 sphalerite, the white lines show a significant left shift of 0.3 eV (Fig. 9G). In contrast to the “origin” (11114.3 eV; no vacancy and Cu^+ substitutions), this substitution of $\text{Ge}^{4+} + \square_{(\text{vacancy})} \rightarrow 2\text{Zn}^{2+}$ (11114.8 eV) results in a relatively smaller shift of white line compared to that of $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 3\text{Zn}^{2+}$ (11114.6 eV; Fig. 9H). The spectral oscillation at around 11118.9 eV gradually increases from C1, through C2, to C3 sphalerite, which is not observed in the calculated spectra (Fig. 9G and H). The presence of vacancies strongly impacts the sphalerite crystal structure, but the substitution of Cu^+ for Zn^{2+} has limited effects due to their similar ionic radii (Cu^+ , 0.75 Å; Zn^{2+} , 0.74 Å; Shannon 1976; Batsanov 2020). In practice, the calculated spectra of “ $\text{Ge}^{4+} + \square_{(\text{vacancy})} \rightarrow 2\text{Zn}^{2+}$ ” and “ $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 3\text{Zn}^{2+}$ ” correspond to the C1 and C3 sphalerite in this study, respectively (Fig. 9G and H). Our study documents that the substitution of $\text{Ge}^{4+} + \square_{(\text{vacancy})} \rightarrow 2\text{Zn}^{2+}$ induces significant shifts of the white line and structural change in sphalerite, compared with $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 3\text{Zn}^{2+}$. Higher Fe contents increase the number of vacancies, thereby favoring the substitution of $\text{Ge}^{4+} + \square_{(\text{vacancy})} \rightarrow 2\text{Zn}^{2+}$, particularly within the C1 sphalerite (Fig. 9G; Wright and Gale 2010).

Implications for Ge enrichment

In Huize deposit, the C1 and C3 are the most economically significant colors for Ge-bearing sphalerite. The former hosts a few Ge-bearing minerals (i.e., argutite), whereas the latter exhibits the highest Ge contents (108–593 ppm, median 386 ppm; Fig. 6). Previous studies have demonstrated that Mn incorporates into sphalerite as MnS, and the Mn contents are significantly and positively correlated with reducing conditions of the environment during sphalerite precipitation (Bernardini et al. 2004; Kelley et al. 2004; Wei et al. 2018; Yu et al. 2019). The sphalerite samples with the highest Ge contents, namely C1 and C3, also exhibit the highest Mn contents. However, there is no distinct linear relationship of their element contents and signals in the LA-ICP-MS mapping, excluding the coupling substitution of Ge and Mn (e.g., $\text{Ge}^{4+} + \text{Mn}^{2+} + \square \leftrightarrow 3\text{Zn}^{2+}$; Figs. 6, 7G and K, and 10F; Hu et al. 2021). This result indicates that the accumulation of Ge in sphalerite is primarily controlled by reducing conditions in the ore-forming fluid.

The Fe/Zn ratio in sphalerite can be utilized to estimate the minimum fluid temperatures during precipitation, but this method is only available for pyrrhotite-free sulfide systems (Keith et al. 2014). Given that Fe content in sphalerite is buffered by pyrite, the FeS activity based on the sphalerite composition can estimate the sulfur fugacity of the ore-forming fluid (Frenzel et al. 2021). The sulfide assemblages associated with the studied sphalerite in stage II consist of pyrite–sphalerite–galena without pyrrhotite, where pyrite serves as the primary Fe-containing sulfide (Figs. 2C–E and 3A, D, and E; Supplementary Fig. 2). Therefore, we can apply these methods to calculate the corresponding ore-forming temperature and sulfur fugacity of different colored sphalerite in this study, and the results are shown in Fig. 11A and B and Supplementary Table 5. However, it should be noted that the GGIMF (Ga, Ge, In, Mn, and Fe) thermometer is unsuitable for the sphalerite in this study due to the presence of certain trace elements below the detection limit in the calculation, particularly Ga and In (Frenzel et al. 2016; Supplementary Table 2).

Based on the calculations, the C1 and C2 sphalerite colors exhibit the highest forming temperatures ranging from 243 to 378 °C and 278 to 308 °C, respectively, followed by the C4 of 244–255 °C and the C5 sphalerite of 242–253 °C (Fig. 11A). Due to their lower Fe contents, the temperatures of C3 and C6 sphalerite are comparatively lower at 227–244 °C and 227–230 °C, respectively. Han et al. (2019) summarized the complete homogenization temperature of fluid inclusions hosted in sphalerite within the Huize Pb–Zn deposit. The results reveal that the sphalerite precipitated at a temperature variation of 150–364 °C. Consequently, the calculated maximum temperature of sphalerite (378 °C in C1 sphalerite) is comparable to the maximum homogenization temperature of fluid inclusions. However, there is a difference of 77 °C between their minimum temperatures (calculated temperature, 227 °C; homogenization temperature, 150 °C). This phenomenon can be attributed to the following factors: firstly, these homogenization temperatures below 227 °C are derived from secondary fluid inclusions; secondly, the Fe/Zn thermometer is not suitable for temperatures below 220 °C (Keith et al. 2014). The C1 and C3 sphalerite colors exhibit the highest sulfur fugacity, followed by the C2 and C6, while the sulfur fugacity of C4 and C5 sphalerite is comparatively lower (Fig. 11B).

Based on these calculation data, two sphalerite models of $\text{C1} \rightarrow \text{C2} \rightarrow \text{C3} \rightarrow \text{C6}$ and $\text{C1} \rightarrow \text{C2} \rightarrow \text{C4} \rightarrow \text{C5} \rightarrow \text{C6}$ both correspond to the descending temperatures (Figs. 11A and 12). The C3 sphalerite hosts the highest Ge contents (median 386 ppm) within a temperature range of 227–244 °C, indicating that sulfur fugacity plays a critical role in controlling Ge enrichment under low-temperature conditions (Figs. 6 and 11B; Frenzel et al. 2021). Meanwhile, Ge tends to accumulate in the C1 sphalerite with higher Fe signals,

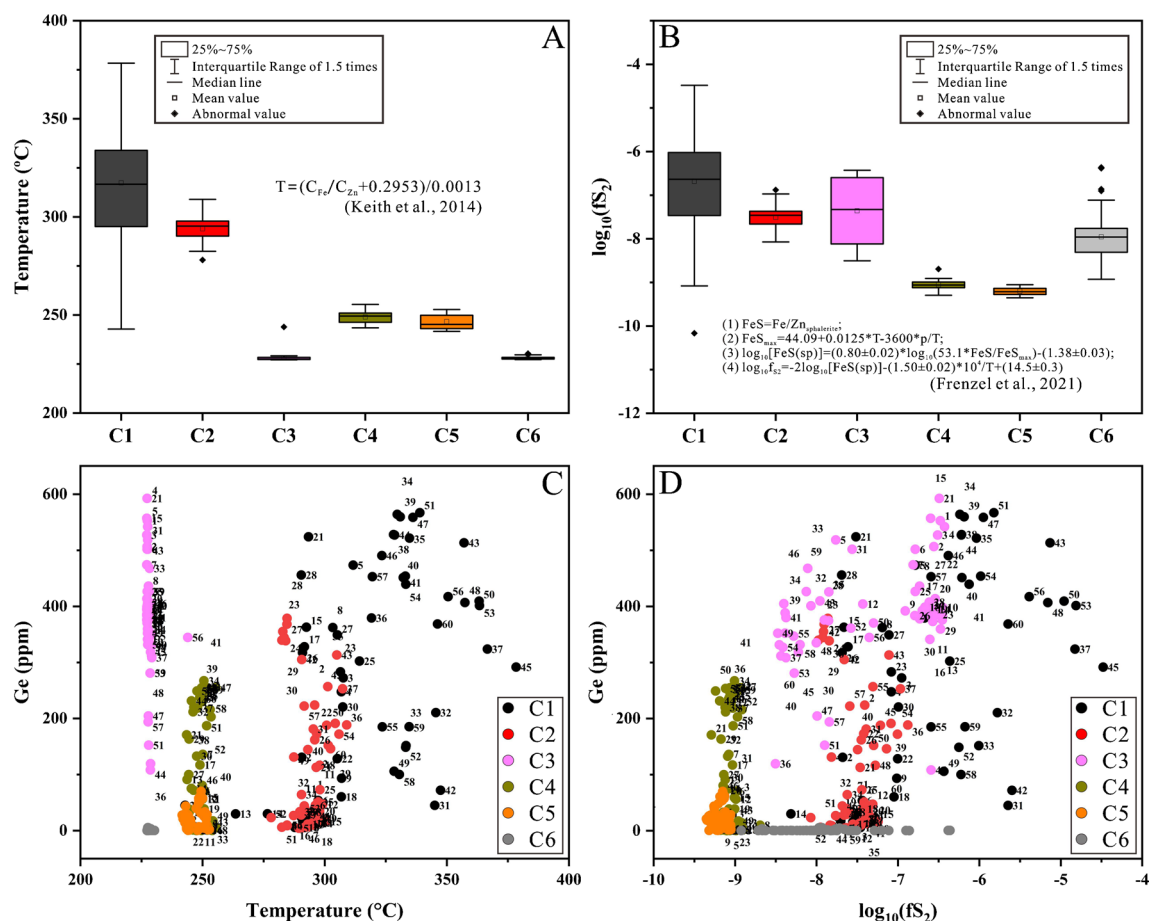
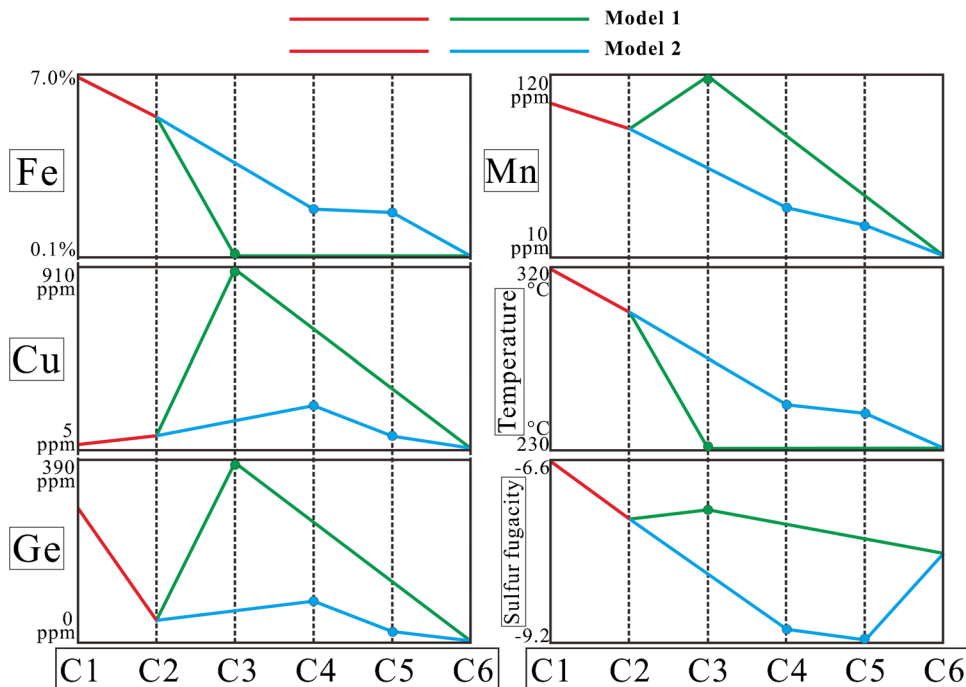


Fig. 11 Histograms of the **A** calculated temperature and **B** sulfur fugacity and **C** and **D** scatter diagrams of Ge content and temperature/sulfur fugacity from the six sphalerite colors in the Huize deposit

(Keith et al. 2014; Frenzel et al. 2021). Pressure ($p = 0.55$ kbar) data from Han et al (2019)

Fig. 12 Relative changes in element contents (Fe, Cu, Ge, and Mn), temperature, and sulfur fugacity [$\log_{10}(fS_2)$] in two models of studied sphalerite. The median value is used to represent each set of data



which also precipitates as fine argutite grains (Fig. 9C and E). This observation indicates that temperature controls the Ge enrichment in the C1 sphalerite (Keith et al. 2014; Liu et al. 2020). However, it is worth noting that these LA-ICP-MS data with higher Ge contents are also associated with elevated temperatures and sulfur fugacity levels in the C1 sphalerite. This implies that temperature and sulfur fugacity jointly regulate the Ge enrichment at high-temperature environments (Fig. 11C and D). Therefore, our findings indicate that sulfur fugacity is the dominant factor controlling Ge enrichment in our studied sphalerite, while temperature may only exert influence at higher temperatures ($> 300\text{ }^{\circ}\text{C}$, Fig. 11C).

As previously discussed, the reducing condition and the sulfur fugacity of the ore-forming fluid contribute to the Ge enrichment in the studied sphalerite (Figs. 6, 10F, 11B, and 12), suggesting that the reduced sulfur content of the ore-forming fluid (e.g., H_2S , HS^- , and S^{2-}) is the leading factor. Mixing of metal-bearing basin brine and reduced sulfur-bearing fluid is the most critical ore-forming mechanism in the Huize deposit (Wang et al. 2016; Meng et al. 2019). As the early stage of fluid mixing and sphalerite precipitation, the C1 sphalerite represents an ore-forming process characterized by heterogeneous physical and chemical conditions, showing the wide temperature and sulfur fugacity ranges (Fig. 11A and B; Kelley et al. 2004; Pfaff et al. 2011; Cugerone et al. 2018). Hence, Ge precipitates either as argutite or incorporates into the C1 sphalerite crystal lattice (i.e., $\text{Ge}^{4+} + \square_{(\text{vacancy})} \rightarrow 2\text{Zn}^{2+}$) with higher temperatures ($> 300\text{ }^{\circ}\text{C}$) and more reduced sulfur contents in the ore-forming fluid (Fig. 9C and D). As fluid mixing progresses, the entire fluid tends to a more stable physicochemical condition and decreasing temperature, and the reduced sulfur content plays a dominant role in precipitating sphalerite with Ge enrichment in the form of $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 3\text{Zn}^{2+}$ (i.e., C2, C3 and C4; Figs. 10D, 11A and B, and 12). In the two sequence sphalerite models, the process of Ge re-enrichment exhibits similarities to that of Cu (e.g., C3 in Model 1, and C2 and C4 in Model 2; Figs. 6 and 12), indicating that further investigation into other factors influencing the incorporation of Cu^+ into sphalerite (i.e., $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 3\text{Zn}^{2+}$) would be beneficial for enhancing Ge enrichment (Belissont et al. 2014; Fougrouse et al. 2023).

Conclusions

In this study, we clarify the Ge distribution and substitution mechanism in the Huize sphalerite and propose possible factors contributing to Ge enrichment. The conclusions are as follows:

- (1) Argutite (GeO_2) and Ge-rich sphalerite are two Ge-bearing phases in the Huize deposit.
- (2) Six sphalerite colors (C1–C6) have been identified, with the C3 sphalerite exhibiting the highest Ge contents (median 386 ppm), followed by the C1 sphalerite (median 288 ppm).
- (3) The presence of Ge in the Huize sphalerite is confirmed through two substitution mechanisms: (1) $\text{Ge}^{4+} + \square_{(\text{vacancy})} \rightarrow 2\text{Zn}^{2+}$, and (2) $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 3\text{Zn}^{2+}$.
- (4) The substitution of $\text{Ge}^{4+} + \square_{(\text{vacancy})} \rightarrow 2\text{Zn}^{2+}$ can significantly impact the sphalerite structure, leading to a shift of white line in the $\mu\text{-XANES}$ spectra, compared to that of $\text{Ge}^{4+} + 2\text{Cu}^+ \rightarrow 3\text{Zn}^{2+}$.
- (5) The Ge enrichment in colored euhedral sphalerite is dominantly controlled by the reduced sulfur content of the ore-forming fluid and temperature during sphalerite precipitation.

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Declarations

Conflict of interest The authors declare no competing interests.

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